Software Requirements Specification

for

Elha2: A Food Saving App for Egypt

Version 1.0 (Milestone 1)

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Revision History

Every sprint adds a row here. Text added or changed in a version is printed in blue with the version in the margin, and removed text is struck through.

Name	Date	Reason For Changes	Version
Elha2 team	09/2026	Phase 0 ideation: problem, market research, proposed solution and technical feasibility	0.1
Elha2 team	10/10/2026	Milestone 1 baseline in the course SRS template: system overview and motivation, full product backlog organised as system features, non-functional requirements and business rules, use case model, global and societal considerations, risks, features to be implemented during the course, and conclusion	1.0

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) specifies the software requirements of Elha2, release 1.0: an Arabic-first, AI-assisted mobile app and web portal that reduces food waste in Egypt. It covers the whole product, not a single subsystem: the household mobile app, the business, NGO, administrator and government web portal, and the backend services they share.

The SRS states *what* Elha2 shall do and how well. *How* it is built is described in the companion Software Design Document (SDD) [3]. This version is the Milestone 1 baseline of the course project. It contains the full product backlog and marks the subset that we will implement during the course (Section 6.6). It is a living document: we update it at the end of every sprint and record each change in the revision history.

1.2 Document Overview

Chapter 1 introduces the product and this document. Chapter 2 describes the product as a whole: its context, functions, users and stakeholders. Chapter 3 specifies the interfaces to users, hardware and other software. Chapter 4 holds the product backlog, organised by feature. Chapter 5 gives the non-functional requirements and business rules. Chapter 6 covers database, localisation and legal requirements, the global and societal considerations, the risks, the features we will implement during the course, and the conclusion. The appendices hold the glossary, the use case model, the open-item list and the verification matrices.

1.3 Product Scope

Elha2 (Egyptian Arabic for "catch it!") is an Arabic-first, AI-assisted mobile app for Android and iOS that reduces food waste in Egypt at three points: **Prevent, Sell, Donate**. A web portal serves food businesses, NGOs, platform administrators and government partners.

- **Prevent (households).** A smart pantry fills itself from receipts, online orders, photos and voice notes. AI estimates when each item will spoil, sends reminders, and suggests Egyptian recipes that use food before it goes off.
- **Sell (businesses).** Restaurants, bakeries, hotels and supermarkets list surplus as surprise bags or specific items at 50–70% off, with a pickup window.
- **Donate (NGOs).** Listings that are not sold or collected pass automatically to a partner NGO; households can also donate sealed pantry items or buy a bag for an NGO.

- **Measure (everyone).** Every confirmed action is converted into kilograms of food saved, Egyptian pounds saved, CO_2e avoided and meals provided, at personal, business, neighbourhood and national level.

1.3.1 Motivation and Benefits

Egyptian households waste about 18 million tonnes of food a year, roughly 163 kg per person against a 79 kg global average, and households produce 60% of all food waste worldwide [4], yet every food-saving app active in Egypt and the Gulf targets business surplus only.

Households are under-served for a simple reason: tracking food at home is tedious, so people stop. Elha2's motivation is therefore twofold. It addresses the largest and most neglected source of waste, and it uses existing AI models to remove the manual effort that makes household tools fail. The benefits are lower grocery bills for households, recovered revenue and lower disposal costs for businesses, a steady flow of safe food for NGOs, and credible data for national reporting. The overall goal is SDG Target 12.3: halving per-capita food waste at retail and consumer level by 2030 [5].

1.3.2 Mission Statement

> *To help Egyptian households and food businesses catch food before it goes to waste, and to turn what remains into affordable meals and donations.*

Every feature is judged against this mission: a requirement that does not serve it is out of scope.

1.3.3 Out of Scope

The first release covers Greater Cairo (Cairo and Giza), in Arabic and English. Elha2 does not run its own delivery fleet: all orders are pickup, and NGOs collect donations themselves or through partner volunteers. Medical or nutritional advice, alcohol, home-cooked food and person-to-person food sharing are out of scope.

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2. Overall Description

2.1 Product Perspective

Elha2 is a new, self-contained product. It sits inside a wider food-saving ecosystem: households and businesses produce surplus, NGOs absorb it, and external services provide login, payments, maps, notifications and AI. Figure 2.1 shows this system boundary as a context diagram; the interfaces are specified in Chapter 3.

The mobile app (Flutter) and the web portal (React) share one backend API (NestJS), which stores data in PostgreSQL with PostGIS (hosted on Supabase) and runs a scheduler for reminders and donation routing. All AI calls pass through a single AI gateway in the backend, except text and barcode recognition, which run on the phone. The architecture, with UML component diagrams, is in the SDD [3]. Figure 2.2 shows how food enters the household pantry without manual typing.

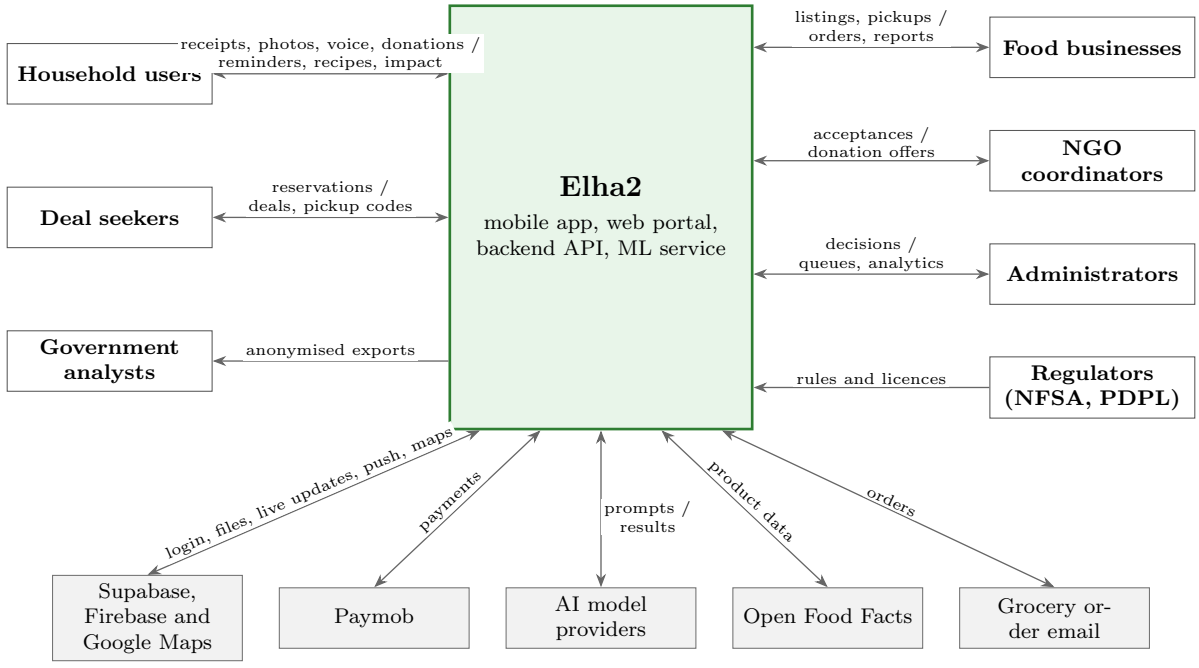


Figure 2.1: Context diagram: the Elha2 system boundary, the people and organisations it serves, and the external systems it depends on. Labels show the data sent in / out.

Five automatic inputs replace typing; manual entry remains only as a fallback. Cooking or donating removes items on its own, and a weekly 30-second swipe check catches the rest, so the pantry stays accurate without daily logging.

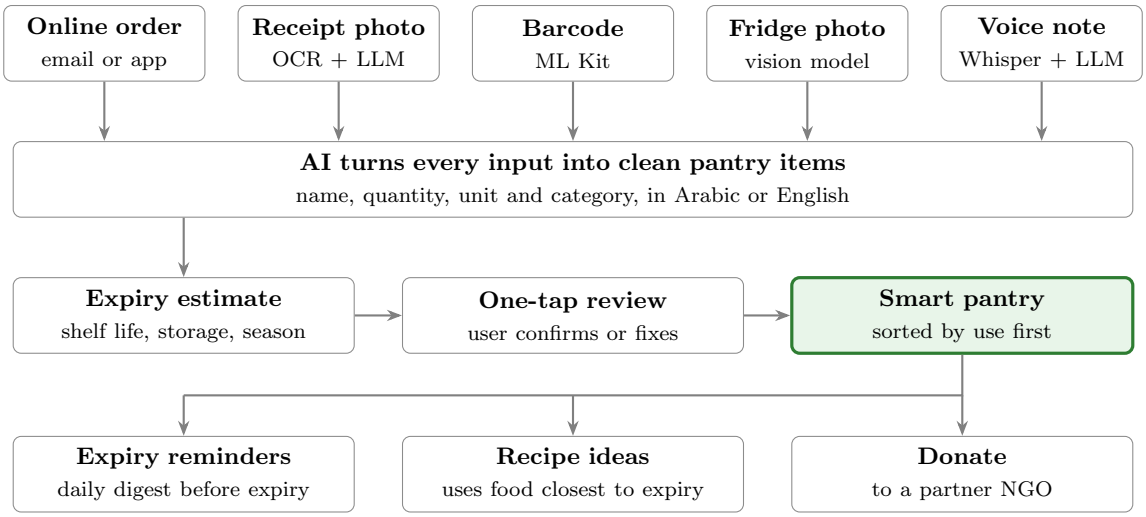


Figure 2.2: Automated pantry pipeline: food enters the pantry with almost no typing.

2.2 Product Functions

Elha2's functions fall into 14 features (epics), E1 to E14. Each one is specified in Chapter 4: feature En is Section 4.n, with its description, priority, stimulus/response sequences and user stories. The user stories together form the product backlog.

ID	Feature	Goal	Stories
E1	Accounts and onboarding	Users and organisations can join quickly and safely	8
E2	Automated smart pantry **[AI]**	Food enters the pantry with almost no typing	11
E3	Expiry intelligence and reminders **[AI]**	Users know what to use first, before it spoils	6
E4	Recipe and cooking assistant **[AI]**	Food near expiry becomes a meal	7
E5	Surplus marketplace	Consumers find and reserve discounted surplus	9
E6	Business portal and listings **[AI]**	Businesses list surplus in under a minute	7
E7	Orders, payments and handover	Reservations are paid and collected reliably	7
E8	NGO donation	Unsold surplus and household donations reach people in need	8
E9	Impact dashboard and reporting	Every user sees credible, verified impact	11
E10	Engagement, Ramadan and events	Habits stick, and peak-waste moments are covered	6
E11	Notifications	The right alert reaches the right user at the right time	5
E12	Trust, safety and moderation	Food and users on the platform can be trusted	4
E13	Administration	Operators can run and configure the platform	6
E14	AI platform (enablers)	AI features are accurate, affordable and monitored	5

ID	Feature	Goal	Stories
	Total		**100**

2.2.1 AI-Supported Functions

AI does the tedious work so that users only confirm. Every AI output is shown with a confidence level, can be corrected in one tap, and has a non-AI fallback.

AI capability	Existing model or service	Job in Elha2
Text and barcode recognition	Google ML Kit (on device)	Reads receipts and barcodes; works offline
Large language model (LLM)	A hosted LLM API such as Claude, GPT or Gemini (**TBD-01**)	Turns messy Arabic and English receipt lines and voice transcripts into clean pantry items; writes recipes; answers "what can I cook?"
Vision-language model	A multimodal LLM	Recognises several items in one photo of a grocery haul or fridge; describes photos of surplus bags to fill in listings
Speech-to-text	Whisper or Google Speech-to-Text (Arabic)	Turns Egyptian Arabic voice notes into text
Recommendation	Collaborative filtering plus distance ranking	Personalised deals and recipe suggestions
Shelf-life estimation	Rules from a food-storage reference (e.g. USDA FoodKeeper), adjusted by an ML model	Predicts expiry dates when none is printed or captured, using storage place and season

2.2.2 Release Plan

The backlog totals 389 story points: 64 Must stories (230 points) form the MVP, 26 Should stories (117 points) form Release 2, and 10 Could stories (42 points) form Release 3 (Figure 2.3).



Figure 2.3: Release roadmap by MoSCoW priority.

The MVP already covers all three pillars (Prevent, Sell, Donate) plus verified impact; later releases add more automatic logging and the heavier AI models. The subset we will implement during the course is defined in Section 6.6.

2.3 User Classes and Characteristics

Users range from tech-savvy students to busy parents with varying literacy, so the household app must work in Egyptian Arabic, with icons and voice input, and without training. Business, NGO and administrator users receive a short onboarding when they join the pilot.

User type	Who they are	What they want	Main features	Platform
Household user (primary)	Parent or young professional managing groceries for a family in Cairo or Giza	Spend less, waste less, without extra effort	Automated pantry, expiry alerts, recipe assistant, donating sealed items, impact dashboard	Mobile app
Deal seeker	University student or budget-conscious worker	Good food at a lower price	Surplus marketplace, map, deal alerts	Mobile app
Business owner or staff	Restaurant, bakery, café, hotel or supermarket	Recover revenue from unsold stock and cut disposal costs	Listing tools, order handover, impact reports	Web portal, staff mobile view
NGO coordinator	Staff or volunteer at a food bank or charity	A steady, predictable flow of safe food	Donation offers, pickup scheduling, beneficiary reporting	Web portal, mobile app
Platform administrator	Elha2 operations team	A safe, trustworthy marketplace	Verification, moderation, disputes, configuration	Web admin console
Government or municipal analyst	Ministry, governorate or research partner	Reliable data on food waste for policy and SDG reporting	Anonymised, aggregated impact data and exports	Web portal (read-only)

The **household user** is the most important class to satisfy: the household pantry gives value even before businesses and NGOs join (risk R-08), and it is where the largest share of waste occurs. Deal seekers, business staff and NGO coordinators are the next most important, because the marketplace and donation loop depend on them. Administrators and government analysts are few in number and are served by the web portal.

2.3.1 Stakeholders

The stakeholders below are the people, organisations and interests affected by Elha2, ranked by the risk of not satisfying them.

Stakeholder	Kind	Interest and risk if not satisfied	Rank
Regulators (NFSA; data protection authority under the PDPL)	Positive	Food safety and personal data; non-compliance could stop the service	High
Household users	Positive (primary users)	Save money and waste less with little effort; without daily use the product fails	High
Food businesses	Positive	Recover revenue from surplus; without listings there is no marketplace	High
NGOs (e.g. the Egyptian Food Bank)	Positive	A steady flow of safe food; without them unsold food has nowhere to go	Medium
Deal seekers	Positive	Affordable food; they absorb the marketplace supply	Medium
Government analysts	Positive	Reliable data for SDG 12.3 reporting	Low
Payment and AI providers	Positive	Use within their terms and rate limits	Low
Business staff who take surplus home informally	Negative	Listing surplus removes an informal benefit, so they may resist	Medium
Informal waste collectors (e.g. Cairo's zabbaleen)	Negative	Part of their income depends on organic waste	Low
Competing food-saving apps	Negative	May copy household features	Low
Food-insecure beneficiaries of NGOs	Voiceless	Receive donated food but do not use the app	Medium
The environment	Voiceless	Emissions and landfill from wasted food	Medium

3. External Interface Requirements

3.1 User Interfaces

Elha2 has two user interfaces: the mobile app and the web portal.

- **Mobile app.** Five main tabs in a bottom navigation bar: Home, Pantry, Deals, Donate and Impact, plus a profile and settings page. The Pantry tab has one large "Add food" button that opens the five capture methods (receipt, barcode, photo, voice, manual).
- **Web portal.** Responsive layout for desktop and tablet, with separate areas for businesses, NGOs, administrators and government analysts, chosen by the signed-in user's role. The counter-staff pickup view also works in a phone browser.
- **Style guides.** Material Design 3 on Android and Apple's Human Interface Guidelines on iOS [10]; one Elha2 colour palette and type scale shared by app and portal.
- **Language and layout.** Arabic by default with a full right-to-left layout; English with left-to-right layout; switchable at any time (US-1.02, NFR-16).
- **Accessibility.** Touch targets of at least 48 dp; system text scaling up to 200%; screen-reader labels on every control; colour is never the only signal (NFR-17). Every main action has an icon and a text label (NFR-18).
- **Standard elements on every screen.** A back control, a help (?) link to the relevant FAQ entry, and an offline banner when the connection is lost.
- **AI results.** Every AI result shows a confidence level and can be edited in one tap before it is saved (NFR-30).
- **Error messages.** Shown inline next to the field or as a snackbar; they say what went wrong and how to fix it, in the user's language, and never show stack traces or codes alone.
- **Confirmations.** Destructive or paid actions (cancel a reservation, delete an account, remove a listing) ask for confirmation; quick changes such as pantry deductions offer a 10 s undo instead (US-2.08).

3.2 Hardware Interfaces

Elha2 needs no special hardware. The mobile app uses the following phone hardware through the operating system's standard APIs, and asks for each permission only when the feature is first used:

Device	Use	Requirements
Camera	Receipts, barcodes, package dates, fridge photos, surplus photos, pickup QR codes	US-2.01, 2.03, 2.05, 3.02, 6.02, 6.04
Microphone	Voice logging in Egyptian Arabic	US-2.06
GPS	Nearby listings and drop-off points, only while the app is open	US-1.04, 5.01; NFR-28
Local storage	Offline pantry and queued changes	NFR-21

Business staff confirm pickups with any phone or laptop camera through the browser. No point-of-sale hardware is required; bulk import from a point-of-sale system uses a CSV export (US-6.06).

3.3 Software Interfaces

External system	Data in and out, and purpose	Interface
Supabase Auth	Out: phone number. In: one-time code result and access token. Phone-number login; the code is sent by SMS through a provider such as Twilio.	Supabase client SDK; token checked on the backend
Supabase Storage	Out: listing photos, organisation documents, photos for AI. In: file links.	Supabase Storage API
Supabase Realtime	In: live listing quantities and order status.	Supabase client SDK (Broadcast channels)
Paymob	Out: amount, order reference. In: payment status, refund status, payout reports. Card and mobile-wallet payments.	REST API, hosted checkout, signed webhooks
Google Maps Platform	Out: addresses and coordinates. In: map tiles, geocodes, directions.	Maps SDK and REST APIs
Firebase Cloud Messaging	Out: device token, notification payload. Push notifications.	FCM HTTP v1 API
Google ML Kit	In: camera image. Out: text lines and barcode values, on the phone.	Mobile SDK (on device)
LLM and vision-language model provider	Out: masked receipt text, voice transcripts, photos, prompts. In: structured JSON items, recipes and tips.	REST API through the AI gateway (US-14.01)
Speech-to-text provider	Out: audio clip. In: Egyptian Arabic transcript.	REST API through the AI gateway
Open Food Facts	Out: barcode. In: product name, brand, category.	REST API
Inbound email service (e.g. SendGrid Inbound Parse)	In: forwarded grocery order emails.	Webhook

Data shared across Elha2's own components (the mobile app, web portal, backend and ML service) passes only through the documented REST API and the internal interfaces specified in the SDD. No component reads another component's database tables directly.

4. System Features

This chapter is the **product backlog**, organised by system feature: one section per feature (epic) E1 to E14. Each section gives the feature's description and priority, its stimulus/response sequences (which correspond to the use cases in Appendix B) and its functional requirements. Each requirement row is one backlog item: a "shall" statement with the user story behind it (in grey), an acceptance test with its verification method (Test, Demonstration, Inspection or Analysis), a MoSCoW priority (Must = MVP, Should = Release 2, Could = Release 3) and story points on the Fibonacci scale. **[AI]** marks a requirement that relies on an AI model.

The backlog holds **100 user stories worth 389 story points**: 64 Must (230 points), 26 Should (117 points) and 10 Could (42 points). Of these, 39 stories (146 points) will be fully implemented during the course (Section 6.6). Non-functional requirements are in Chapter 5.

ID	Feature	Priority	Stories	Must	Should	Could	Points
E1	Section 4.1: Accounts and Onboarding	High	8	7	1	0	27
E2	Section 4.2: Automated Smart Pantry **[AI]**	High	11	8	3	0	50
E3	Section 4.3: Expiry Intelligence and Reminders **[AI]**	High	6	4	1	1	22
E4	Section 4.4: Recipe and Cooking Assistant **[AI]**	Medium	7	3	2	2	23
E5	Section 4.5: Surplus Marketplace	High	9	6	2	1	30
E6	Section 4.6: Business Portal and Listings **[AI]**	High	7	5	1	1	31
E7	Section 4.7: Orders, Payments and Handover	High	7	6	1	0	29
E8	Section 4.8: NGO Donation	High	8	4	3	1	35
E9	Section 4.9: Impact Dashboard and Reporting	High	11	5	5	1	49
E10	Section 4.10: Engagement, Ramadan and Events	Low	6	0	3	3	25
E11	Section 4.11: Notifications	Medium	5	4	1	0	12
E12	Section 4.12: Trust, Safety and Moderation	High	4	4	0	0	12
E13	Section 4.13: Administration	Medium	6	4	2	0	22
E14	Section 4.14: AI Platform (Enablers)	High	5	4	1	0	22
	Total		**100**	**64**	**26**	**10**	**389**

4.1 E1 Accounts and Onboarding

4.1.1 Description and Priority

Lets households, deal seekers, businesses and NGOs join Elha2 quickly and safely: phone-number sign-up, language choice, location, household sharing, organisation registration and account deletion. Every other feature depends on it, so it is built first.

The feature has 8 stories worth 27 points (7 Must, 1 Should, 0 Could). Built during the course: US-1.01 (Sprint 1), US-1.02 (Sprint 1), US-1.04 (Sprint 1), US-1.06 (Sprint 1), US-1.07 (Sprint 1).

Priority: High. Benefit 8, penalty 9, cost 4, risk 2 (1 = low, 9 = high).

4.1.2 Stimulus/Response Sequences

Related use cases: UC-37, UC-38, UC-42 (Appendix B).

#	Stimulus (user or system action)	Response (system behaviour)
1	A new user enters an Egyptian mobile number and taps *Send code*.	The system sends an SMS one-time code within 60 s and shows the code screen.
2	The user enters a valid code.	The system creates the account, signs the user in and opens the short onboarding.
3	The user enters a wrong code 5 times within an hour.	The system blocks further attempts for that number for 1 h and says when to try again.
4	A business owner submits the registration form with licence documents.	The system creates the organisation in *pending* state and tells the owner it is awaiting review.
5	A user taps *Delete account* and confirms.	The system signs the user out and schedules deletion of personal data within 30 days.

4.1.3 Functional Requirements

ID	Requirement and user story	Acceptance test	Priority	Pts
US-1.01	**Sign up with a phone number** The system shall register a new user with an Egyptian mobile number and an SMS one-time code delivered within 60 s. *As a new user, I want to sign up with my Egyptian mobile number and a one-time code so that I can join without a password.*	**Test.** Code arrives by SMS within 60 s; expires after 5 min; at most 5 attempts per hour per number.	Must	3

ID	Requirement and user story	Acceptance test	Priority	Pts
US-1.02	**Switch the app language** The system shall display every screen in Arabic (default) or English, with a right-to-left layout for Arabic. *As a user, I want to use the app in Arabic (default) or English so that I can read it comfortably.*	**Demonstration.** Language switch applies instantly; Arabic uses a full right-to-left layout and Arabic numerals as an option.	Must	3
US-1.03	**Complete a short onboarding** The system shall collect household size and dietary needs in an optional onboarding of at most 3 screens. *As a new household user, I want a short onboarding that asks my household size and dietary needs so that reminders and recipes fit my home.*	**Demonstration.** At most 3 screens; can be skipped; answers editable later in settings.	Must	2
US-1.04	**Set a location** The system shall store the user's location from GPS or a chosen area and show other users only an approximate distance. *As a user, I want to set my location by GPS or by choosing my area so that I see nearby deals.*	**Test.** My exact location is never shown to other users, only an approximate distance.	Must	3
US-1.05	**Share a household pantry** The system shall let a household owner invite members by link to one shared pantry that syncs changes within 5 s. *As a household user, I want to invite family members to one shared pantry so that everyone sees and updates the same food.*	**Test.** Invite by link; owner and member roles; changes appear on other phones within 5 s.	Should	5
US-1.06	**Register a business** The system shall register a business and its branches with a commercial registration and food-safety licence and keep the account pending until an administrator approves it. *As a business owner, I want to register my business and its branches with my commercial registration and food-safety licence so that I can list surplus.*	**Demonstration.** Documents uploaded in the portal; account stays "pending" until an admin approves; multiple branches per business.	Must	5
US-1.07	**Register an NGO** The system shall register an NGO with its official registration number and block donation offers until an administrator verifies it. *As an NGO coordinator, I want to register my organisation with its official registration number so that we can receive donations.*	**Demonstration.** Admin verification before first donation; several coordinators per NGO.	Must	3

ID	Requirement and user story	Acceptance test	Priority	Pts
US-1.08	**Delete an account** The system shall delete a user's personal data within 30 days of an account deletion request. *As a user, I want to edit my profile, control my privacy settings and delete my account so that I stay in control of my data.*	**Test.** Profile and privacy settings are editable; deletion removes personal data within 30 days; anonymised impact totals are kept.	Must	3

4.2 E2 Automated Smart Pantry [AI]

4.2.1 Description and Priority

Keeps a household's food inventory with almost no typing. Items enter from receipts, barcodes, online orders, photos and voice notes, are reviewed in one tap, and leave the pantry automatically when they are cooked or donated. Typing is the last resort, and the pantry corrects itself through a weekly 30-second check rather than constant logging.

The feature has 11 stories worth 50 points (8 Must, 3 Should, 0 Could). Built during the course: US-2.01 (Sprint 1), US-2.02 (Sprint 1), US-2.07 (Sprint 1), US-2.08 (Sprint 2), US-2.09 (Sprint 2), US-2.10 (Sprint 1), US-2.11 (Sprint 1).

Priority: High. Benefit 9, penalty 8, cost 8, risk 8 (1 = low, 9 = high).

4.2.2 Stimulus/Response Sequences

Related use cases: UC-01 to UC-09, UC-15, UC-16 (Appendix B).

#	Stimulus (user or system action)	Response (system behaviour)
1	The household user photographs a paper receipt.	The app reads the text on the phone, masks personal data, and within 10 s shows the extracted items with confidence levels.
2	The user edits a low-confidence item and taps *Confirm all*.	The system saves all items to the household pantry, each with an expiry date.
3	The AI service times out or returns invalid data.	The app shows the recognised text lines and lets the user add items manually.
4	The user scans a barcode while offline.	The app reads the code on the phone, fills in known products from the local cache, and queues the item for sync.
5	The user taps *I cooked this* on a recipe.	The system deducts the ingredients from the pantry and offers *Undo* for 10 s.
6	The weekly check is due.	The app shows at most 10 cards for the most uncertain items; each swipe marks an item used, thrown away or still here.

4.2.3 Functional Requirements

ID	Requirement and user story	Acceptance test	Priority	Pts
US-2.01	**Scan a receipt [AI]** The system shall add the items on a photographed paper receipt to the pantry, reading at least 90% of item lines correctly within 10 s. *As a household user, I want to photograph a paper receipt and have every item added at once so that I never type my shopping.*	**Test.** On-device OCR reads the receipt; an LLM turns lines into name, quantity, unit and category, handling mixed Arabic and English and shop abbreviations; at least 90% of item lines correct on our test set; result in under 10 s.	Must	8
US-2.02	**Review AI-extracted items [AI]** The system shall show every AI-extracted item with a confidence level for confirmation before it is saved. *As a household user, I want to review what the AI found before it is saved so that mistakes do not enter my pantry.*	**Demonstration.** Each item shows a confidence level; low-confidence items are highlighted; edit or delete in one tap; "Confirm all" saves everything.	Must	3
US-2.03	**Scan a barcode [AI]** The system shall fill in a packaged item's details from its barcode within 2 s, including when offline. *As a household user, I want to scan a barcode to add a packaged item so that its name and details fill in automatically.*	**Test.** ML Kit scans on the phone in under 2 s, also offline; product looked up in our catalogue and an open product database; unknown barcodes saved for admin review.	Must	3
US-2.04	**Import online grocery orders [AI]** The system shall add the items from a forwarded grocery order email to the pantry within 2 min of receipt. *As a household user, I want to forward my online grocery order emails, or link a supported grocery app, so that deliveries are added without any action from me.*	**Test.** Each user has a private forwarding address; an LLM parses order confirmations from Egyptian grocery apps (e.g. Breadfast, Talabat Mart); items appear within 2 min.	Should	8
US-2.05	**Photograph fridge or groceries [AI]** The system shall recognise the items and counts in one photo of groceries or an open fridge and send them to the review screen. *As a household user, I want to take one photo of my grocery bags or open fridge so that several items are recognised at once.*	**Analysis.** A vision-language model lists the items it sees with counts; I confirm on the review screen; the photo is deleted after processing unless I opt in to help improve the model.	Should	8

ID	Requirement and user story	Acceptance test	Priority	Pts
US-2.06	**Log items by voice [AI]** The system shall extract items and quantities from an Egyptian Arabic voice note and send them to the review screen. *As a household user, I want to say what I bought in Egyptian Arabic (e.g. "I bought a kilo of tomatoes and 5 loaves of bread") so that I can log food hands-free.*	**Analysis.** Speech-to-text plus LLM extract items and quantities; works with Egyptian dialect; results go to the review screen.	Should	5
US-2.07	**Add an item manually** The system shall let a user add a pantry item by hand in 3 taps or fewer, with Arabic and English autocomplete. *As a household user, I want to add an item by hand with smart autocomplete so that it is quick when nothing else fits.*	**Demonstration.** Suggestions in Arabic and English, including common Egyptian foods, with a default quantity; an item is added in 3 taps or fewer.	Must	2
US-2.08	**Update pantry automatically** The system shall reduce pantry quantities automatically when a recipe is cooked or an item is donated, with a 10 s undo. *As a household user, I want my pantry to update itself when I cook a suggested recipe or donate an item so that I do not have to log usage.*	**Test.** Quantities are reduced automatically; an "Undo" is available for 10 s.	Must	3
US-2.09	**Complete the weekly check [AI]** The system shall present a weekly check of at most 10 items, chosen as the most uncertain, to mark as used, thrown away or still here. *As a household user, I want a weekly 30-second check where I swipe items as used, thrown away or still here so that the pantry stays accurate without daily effort.*	**Test.** At most 10 cards, chosen by the model as the most uncertain items; items well past their expected use are archived automatically after a grace period.	Must	5
US-2.10	**Record thrown-away food** The system shall record a reason from a fixed list whenever an item is marked as thrown away. *As a household user, I want to mark an item as thrown away with a reason (spoiled, forgotten, cooked too much) so that my real waste is measured.*	**Test.** Fixed reason list plus "other"; data feeds the waste rate in E9.	Must	2
US-2.11	**View the pantry** The system shall show the pantry grouped by storage place and sorted by expiry date, colour-coded for today, within 3 days and later. *As a household user, I want to see my pantry grouped by fridge, freezer and cupboard and sorted by "use first" so that I know what to eat next.*	**Demonstration.** Search and filter; colour code for expires today, within 3 days, later.	Must	3

4.3 E3 Expiry Intelligence and Reminders [AI]

4.3.1 Description and Priority

Gives every pantry item an expiry date, printed or estimated, and reminds the household before food spoils. It explains the difference between "use by" and "best before" so that safe food is not thrown away.

The feature has 6 stories worth 22 points (4 Must, 1 Should, 1 Could). Built during the course: US-3.01 (Sprint 2), US-3.03 (Sprint 2), US-3.04 (Sprint 2).

Priority: High. Benefit 8, penalty 7, cost 5, risk 5 (1 = low, 9 = high).

4.3.2 Stimulus/Response Sequences

Related use cases: UC-10, UC-11 (Appendix B).

#	Stimulus (user or system action)	Response (system behaviour)
1	An item is saved without a printed date.	The system estimates an expiry date from the shelf-life reference, storage place and season, and marks it *estimated*.
2	The user's chosen digest time arrives and items expire within the lead time.	The system sends one push digest listing those items.
3	The user opens an expiry notification.	The app opens the item with recipe and donate actions.

4.3.3 Functional Requirements

ID	Requirement and user story	Acceptance test	Priority	Pts
US-3.01	**Estimate an expiry date [AI]** The system shall estimate an expiry date, marked as estimated, for every item saved without a printed date. *As a household user, I want the app to estimate an expiry date when none was captured so that every item gets a reminder.*	**Test.** Estimate from a food-storage reference adjusted by storage place and season (shorter in summer); shown as "estimated"; I can override it.	Must	5
US-3.02	**Read a printed date [AI]** The system shall read the printed date and its label type (use by or best before) from a package photo. *As a household user, I want to photograph the printed date on a package so that the exact date is saved.*	**Analysis.** OCR reads common date formats, including Arabic numerals; detects whether it is "use by" or "best before".	Should	5

ID	Requirement and user story	Acceptance test	Priority	Pts
US-3.03	**Receive a daily expiry digest** The system shall send at most one daily digest listing items that expire within the user's chosen lead time. *As a household user, I want one daily digest of items expiring in the next 1–2 days so that I am reminded without being spammed.*	**Test.** I choose the time and the lead time; at most one digest per day, plus same-day alerts for "use by" items.	Must	3
US-3.04	**Explain date labels** The system shall show each item's label type with a one-line explanation of use by and best before in Egyptian Arabic. *As a household user, I want "use by" and "best before" explained in plain Egyptian Arabic so that I stop throwing away safe food.*	**Inspection.** Label type shown on each item with a one-line explanation.	Must	1
US-3.05	**Show at-risk items on the home screen** The system shall provide an Android and iOS home-screen widget showing the 3 items closest to expiry. *As a household user, I want a home-screen widget with my 3 most at-risk items so that I see them without opening the app.*	**Demonstration.** Android and iOS widgets; tapping opens the item.	Could	5
US-3.06	**Act on an expiring item** The system shall offer recipe and donate actions directly from an expiry notification and from the item page. *As a household user, I want one-tap actions on an expiring item (find a recipe or donate it) so that I can act straight from the reminder.*	**Demonstration.** Actions available from the notification and the item page.	Must	3

4.4 E4 Recipe and Cooking Assistant [AI]

4.4.1 Description and Priority

Turns food that is close to expiry into a meal. Suggestions come first from a curated Egyptian recipe database, ranked by how many at-risk items they use, with LLM-generated recipes as a checked fallback. Cooking a recipe updates the pantry and the user's impact.

The feature has 7 stories worth 23 points (3 Must, 2 Should, 2 Could). Built during the course: US-4.01 (Sprint 2), US-4.05 (Sprint 2).

Priority: Medium. Benefit 6, penalty 5, cost 5, risk 5 (1 = low, 9 = high).

4.4.2 Stimulus/Response Sequences

Related use cases: UC-12 to UC-14 (Appendix B).

#	Stimulus (user or system action)	Response (system behaviour)
1	The user opens *Recipes* from the pantry.	The system shows the 5 recipes that use the most items closest to expiry.
2	The user applies meal, time, budget or diet filters.	The list updates instantly with matching recipes.
3	The user asks the cooking assistant a question.	The system answers in the user's language using the pantry, with a reheating safety note for leftovers, and declines non-food topics.
4	The user taps *I cooked this.*	The system deducts the ingredients and adds the food saved to the user's impact.

4.4.3 Functional Requirements

ID	Requirement and user story	Acceptance test	Priority	Pts
US-4.01	**Suggest recipes for at-risk food [AI]** The system shall suggest the 5 Egyptian recipes that use the most items closest to expiry. *As a household user, I want Egyptian recipe suggestions that use my items closest to expiry so that they become a meal, not waste.*	**Test.** Top 5 ranked by how many at-risk items they use; drawn from a curated Egyptian recipe database first, with LLM-generated recipes as a fallback that pass a food-safety check.	Must	5
US-4.02	**Ask the cooking assistant [AI]** The system shall answer cooking questions in Arabic or English using the user's pantry and include a reheating safety note for leftovers. *As a household user, I want to ask in chat "what can I cook with leftover rice and chicken?" so that I get ideas in my own words.*	**Analysis.** Answers in Arabic or English using my pantry; declines non-food topics; adds a reheating safety note for leftovers.	Should	5
US-4.03	**Filter recipes** The system shall filter recipe suggestions by meal, preparation time, budget and diet. *As a household user, I want to filter recipes by meal (breakfast, lunch, dinner, iftar, suhoor), time, budget and diet so that suggestions fit my day.*	**Test.** Filters combine; results update instantly.	Must	3

ID	Requirement and user story	Acceptance test	Priority	Pts
US-4.04	**Scale a recipe [AI]** The system shall scale recipe quantities to the household size and suggest substitutes for missing ingredients. *As a household user, I want a recipe scaled to my household and adjusted to what I have so that I cook the right amount.*	**Test.** Quantities scale with household size; missing ingredients get suggested substitutes.	Should	3
US-4.05	**Mark a meal as cooked** The system shall deduct a cooked recipe's ingredients from the pantry and add the food saved to the user's impact. *As a household user, I want to tap "I cooked this" so that ingredients leave my pantry and the food saved is counted.*	**Test.** Ingredients deducted (see US-2.08); kg saved added to my impact.	Must	2
US-4.06	**Save and share recipes** The system shall let users save favourite recipes and share them on WhatsApp as an image card. *As a household user, I want to save favourite recipes and share them on WhatsApp so that I can reuse and spread them.*	**Demonstration.** Favourites list; share as an image card with the Elha2 link.	Could	2
US-4.07	**Get leftover ideas [AI]** The system shall suggest uses for a logged leftover on the following morning. *As a household user, I want ideas for leftovers of a cooked meal the next day, such as stale bread into fattah, so that leftovers are eaten.*	**Test.** Logging a leftover triggers suggestions the next morning.	Could	3

4.5 E5 Surplus Marketplace

4.5.1 Description and Priority

Lets consumers find, reserve and rate discounted surplus from verified food businesses near them. It is the demand side of the Sell pillar.

The feature has 9 stories worth 30 points (6 Must, 2 Should, 1 Could). Built during the course: US-5.01 (Sprint 2), US-5.03 (Sprint 2), US-5.04 (Sprint 2), US-5.07 (Sprint 2).

Priority: High. Benefit 8, penalty 7, cost 5, risk 3 (1 = low, 9 = high).

4.5.2 Stimulus/Response Sequences

Related use cases: UC-20 to UC-24 (Appendix B).

#	Stimulus (user or system action)	Response (system behaviour)
1	The deal seeker opens the Deals tab.	The system shows listings within the chosen radius on a map and in a list, sorted by distance, within 2 s.
2	The deal seeker taps *Reserve* on a listing with stock left.	The system reduces the quantity in one transaction and confirms the reservation.
3	Two users try to reserve the last bag at the same time.	Exactly one reservation succeeds; the other user sees *sold out*.
4	The buyer reports a problem with a photo within 24 h of pickup.	The system opens a dispute in the administrator queue and acknowledges the report.

4.5.3 Functional Requirements

ID	Requirement and user story	Acceptance test	Priority	Pts
US-5.01	**Browse nearby surplus** The system shall list surplus within a chosen radius of 1 to 20 km on a map and in a list within 2 s. *As a deal seeker, I want to browse nearby surplus listings on a map and in a list so that I find food close to me.*	**Test.** Radius from 1 to 20 km; results load in under 2 s; list sorted by distance by default.	Must	5
US-5.02	**Filter listings** The system shall filter listings by category, price, pickup time, distance and dietary needs and remember the filters between sessions. *As a deal seeker, I want to filter by category, price, pickup time, distance and dietary needs so that I only see what suits me.*	**Test.** Filters combine and are remembered between sessions.	Must	3
US-5.03	**View a listing** The system shall show each listing's original and discounted price, pickup window, allergens, photo, rating and live quantity. *As a deal seeker, I want a listing page with original and discounted price, pickup window, allergens, photo, rating and quantity left so that I can decide quickly.*	**Demonstration.** All fields shown; quantity updates live.	Must	2
US-5.04	**Reserve a bag** The system shall reserve a bag in a single transaction so that no bag is reserved twice. *As a deal seeker, I want to reserve a bag so that it is held for me until pickup.*	**Test.** Quantity decreases in one transaction, so a bag is never sold twice; reservation confirmed on screen and by notification.	Must	5

ID	Requirement and user story	Acceptance test	Priority	Pts
US-5.05	**Follow a store** The system shall alert a store's followers within 1 min of a new listing. *As a deal seeker, I want to follow favourite stores so that I am alerted when they list.*	**Test.** Follow from the store page; alert within 1 min of a new listing.	Should	3
US-5.06	**See a personal feed [AI]** The system shall rank deals for each user by past orders, favourites, distance and time of day. *As a deal seeker, I want a personalised "For you" feed so that the most relevant deals appear first.*	**Analysis.** Ranking uses my past orders, favourites, distance and time of day; works with a default ranking for new users.	Could	5
US-5.07	**Get directions** The system shall open Google Maps with directions to the store. *As a deal seeker, I want directions to the store so that I can get there easily.*	**Demonstration.** Opens Google Maps with the store location.	Must	1
US-5.08	**Rate a store** The system shall let only buyers with a confirmed pickup rate a store from 1 to 5 stars. *As a deal seeker, I want to rate and review a store after pickup so that others know what to expect.*	**Test.** Only verified buyers, after a confirmed pickup; 1–5 stars plus optional text.	Should	3
US-5.09	**Report a problem with an order** The system shall accept a problem report with a photo within 24 h of pickup and send it to the dispute queue. *As a deal seeker, I want to report a problem with an order, with a photo, so that unsafe or wrong food is dealt with.*	**Test.** Report within 24 h of pickup; goes to the admin dispute queue (US-13.03).	Must	3

4.6 E6 Business Portal and Listings [AI]

4.6.1 Description and Priority

Gives food businesses the tools to list surplus in under a minute, hand over orders by code and manage staff. It is the supply side of the Sell pillar.

The feature has 7 stories worth 31 points (5 Must, 1 Should, 1 Could). Built during the course: US-6.01 (Sprint 2), US-6.04 (Sprint 3), US-6.07 (Sprint 3).

Priority: High. Benefit 8, penalty 8, cost 6, risk 4 (1 = low, 9 = high).

4.6.2 Stimulus/Response Sequences

Related use cases: UC-25, UC-27, UC-28, UC-29, UC-30, UC-31, UC-40 (Appendix B).

#	Stimulus (user or system action)	Response (system behaviour)
1	Staff fill in the listing form and tap *Publish*.	The system validates price (discount 30–80%) and allergens, then publishes the listing and alerts followers.
2	Staff take a photo of the surplus.	The system suggests title, category, weight and likely allergens for staff to confirm.
3	Counter staff scan a customer's QR code.	The system checks the code belongs to an open order at this branch and marks it collected.
4	Staff cancel a listing that has reservations.	The system notifies every buyer and starts their refunds automatically.

4.6.3 Functional Requirements

ID	Requirement and user story	Acceptance test	Priority	Pts
US-6.01	**Create a listing** The system shall publish a listing with price, original value, quantity, pickup window, allergens and photo, accepting only discounts between 30% and 80%. *As a business staff member, I want to create a listing (surprise bag or specific item) with price, original value, quantity, pickup window, allergens and photo so that customers can reserve it.*	**Test.** Done in under 60 s; discount must be between 30% and 80%; allergens required.	Must	5
US-6.02	**Fill a listing from a photo [AI]** The system shall suggest a listing's title, description, category, weight and likely allergens from a photo for staff to confirm. *As a business staff member, I want to take a photo of the surplus and have the listing filled in for me so that listing takes seconds.*	**Analysis.** A vision-language model suggests title, description, category, estimated weight and likely allergens; staff confirm before publishing.	Should	5
US-6.03	**Manage staff roles** The system shall restrict counter staff to viewing the day's orders and confirming pickups. *As a business owner, I want to manage branches and staff accounts with roles (owner, manager, counter staff) so that each person sees only what they need.*	**Test.** Counter staff can only view today's orders and confirm pickups.	Must	5
US-6.04	**Confirm a pickup** The system shall confirm a pickup from a scanned QR code or a typed 4-digit code in any phone browser. *As counter staff, I want to see today's reservations and confirm pickup by scanning the customer's code so that handover is fast.*	**Test.** Scan or type the 4-digit code; works on any phone browser.	Must	3

ID	Requirement and user story	Acceptance test	Priority	Pts
US-6.05	**Edit or cancel a listing** The system shall notify and refund every customer with a reservation when a listing is cancelled. *As a business staff member, I want to edit or cancel a listing so that I can react to what is left.*	**Test.** Customers with reservations are notified and refunded automatically (US-7.05).	Must	3
US-6.06	**Import stock in bulk** The system shall create listings from a CSV file after validating and previewing every row. *As a business manager, I want to import unsold stock from a POS or CSV export at closing so that listing happens in bulk.*	**Test.** CSV template provided; rows validated and previewed before publishing.	Could	8
US-6.07	**Turn on auto-donation** The system shall store a default donation partner and an auto-donation setting for each branch. *As a business owner, I want to choose a default donation partner and turn on auto-donation so that nothing unsold is wasted.*	**Test.** Setting per branch; used by US-8.02.	Must	2

4.7 E7 Orders, Payments and Handover

4.7.1 Description and Priority

Handles payment, pickup codes, cancellations, refunds, order history and weekly payouts so that reservations are paid and collected reliably. During the course, cash at pickup replaces in-app payment because a live payment account requires a registered company (R-04).

The feature has 7 stories worth 29 points (6 Must, 1 Should, 0 Could). Built during the course: US-7.02 (Sprint 2), US-7.03 (Sprint 2).

Priority: High. Benefit 8, penalty 8, cost 7, risk 6 (1 = low, 9 = high).

4.7.2 Stimulus/Response Sequences

Related use cases: UC-22, UC-23, UC-25, UC-26 (Appendix B).

#	Stimulus (user or system action)	Response (system behaviour)
1	The buyer chooses card or wallet payment.	The system opens Paymob's hosted checkout and confirms the order within 5 s of payment.
2	The buyer cancels at least 1 h before the pickup window.	The system cancels the order, releases the bag and refunds the buyer in full.
3	The pickup window ends without collection.	The system records a no-show and, after 3 no-shows in 30 days, blocks new reservations for 7 days.

#	Stimulus (user or system action)	Response (system behaviour)
4	The weekly payout job runs.	The system pays each business net of commission and makes the statement available as PDF and CSV.

4.7.3 Functional Requirements

ID	Requirement and user story	Acceptance test	Priority	Pts
US-7.01	**Pay in the app** The system shall take card and mobile-wallet payments through Paymob and confirm them within 5 s without storing card numbers. *As a deal seeker, I want to pay in the app by card or mobile wallet so that my reservation is secured.*	**Test.** Payment through Paymob; Elha2 never stores card numbers; confirmation in under 5 s.	Must	8
US-7.02	**Pay cash at pickup** The system shall allow cash at pickup only for users with at least 2 completed pickups and no recent no-shows. *As a trusted deal seeker, I want to pay cash at pickup so that I can order without a card.*	**Test.** Only for users with at least 2 completed pickups and no recent no-shows.	Should	3
US-7.03	**Show a pickup code** The system shall give every order a QR code and a 4-digit code that mark it collected when scanned. *As a deal seeker, I want a QR code and 4-digit code for my order so that pickup is quick and confirmed.*	**Test.** Scanning marks the order collected and records it as verified impact (US-9.03).	Must	5
US-7.04	**Cancel a reservation** The system shall refund a reservation cancelled at least 1 h before its pickup window and block reservations for 7 days after 3 no-shows in 30 days. *As a deal seeker, I want to cancel a reservation before the pickup window so that I get my money back.*	**Test.** Full refund if cancelled at least 1 h before the window (configurable); 3 no-shows in 30 days block reservations for 7 days.	Must	3
US-7.05	**Refund a business cancellation** The system shall start a refund immediately when a business cancels an order and notify the buyer with the reason. *As a deal seeker, I want an automatic refund when a business cancels so that I am never out of pocket.*	**Test.** Refund started immediately; I am notified with the reason.	Must	3

ID	Requirement and user story	Acceptance test	Priority	Pts
US-7.06	**View order history** The system shall keep an order history with e-receipts showing original value, price paid and money saved. *As a user, I want my order history and e-receipts so that I can track my spending.*	**Demonstration.** Receipts show original value, price paid and money saved.	Must	2
US-7.07	**Receive payouts** The system shall pay businesses weekly with a statement, net of commission, downloadable as PDF and CSV. *As a business owner, I want weekly payouts with a clear statement so that I know what I earned.*	**Test.** Commission deducted; statement downloadable as PDF and CSV.	Must	5

4.8 E8 NGO Donation

4.8.1 Description and Priority

Sends food that is not sold or collected to verified NGOs that can handle it, and closes the loop by recording meals distributed. It also lets households donate sealed pantry items at partner drop-off points and buy bags for NGOs.

The feature has 8 stories worth 35 points (4 Must, 3 Should, 1 Could). Built during the course: US-8.01 (Sprint 3), US-8.02 (Sprint 3), US-8.04 (Sprint 3).

Priority: High. Benefit 8, penalty 7, cost 5, risk 5 (1 = low, 9 = high).

4.8.2 Stimulus/Response Sequences

Related use cases: UC-32 to UC-36 (Appendix B).

#	Stimulus (user or system action)	Response (system behaviour)
1	A listing's pickup window ends with unsold quantity and auto-donation on.	The scheduler offers the food to the best-matched NGO.
2	The NGO does not answer within 30 min or declines.	The system offers the food to the next matching NGO.
3	The NGO coordinator accepts and books a pickup time.	The system notifies the business of the booked time.
4	The NGO records meals distributed for a donation.	The system stores the number, without any beneficiary data, and updates impact.

4.8.3 Functional Requirements

ID	Requirement and user story	Acceptance test	Priority	Pts
US-8.01	**Set donation preferences** The system shall store each NGO's service area, accepted food types, capacity and pickup hours for matching. *As an NGO coordinator, I want to set our service area, accepted food types, capacity and pickup hours so that we only get offers we can handle.*	**Test.** Settings editable at any time; used for matching.	Must	3
US-8.02	**Route unsold food to an NGO** The system shall offer unsold food to the best-matched NGO when a pickup window ends and to the next NGO after 30 min without acceptance. *As a business owner, I want unsold items offered automatically to a matched NGO when the pickup window ends so that they are not thrown away.*	**Test.** Scheduler finds NGOs by distance, food type and capacity; first NGO has 30 min to accept, then the next is offered.	Must	8
US-8.03	**Plan a collection route [AI]** The system shall suggest a collection route for several donations ordered by pickup window and distance. *As an NGO coordinator, I want a suggested collection route for several donations so that one trip collects more food.*	**Test.** Route ordered by pickup windows and distance; opens in Google Maps.	Could	8
US-8.04	**Accept a donation** The system shall let an NGO accept or decline an offer, book a pickup time and notify the business. *As an NGO coordinator, I want to accept or decline an offer and book a pickup time so that the business knows when we will come.*	**Test.** Business notified on acceptance; pickup confirmed by code as in US-7.03.	Must	3
US-8.05	**Donate a bag** The system shall send a bag bought for donation to a matched NGO and add it to the buyer's impact. *As a user, I want to buy a bag and donate it so that someone in need receives a meal.*	**Test.** Donated bags go to a matched NGO; I see it in my impact.	Should	5
US-8.06	**Donate pantry items** The system shall show partner drop-off points on a map and record a donation once the NGO confirms it. *As a household user, I want to donate sealed pantry items at partner drop-off points so that food I will not use still helps.*	**Demonstration.** Map of drop-off points; donation confirmed by the NGO.	Should	3

ID	Requirement and user story	Acceptance test	Priority	Pts
US-8.07	**Record meals distributed** The system shall record the number of meals distributed per donation without storing beneficiary data. *As an NGO coordinator, I want to record how many meals we distributed from each donation so that the loop is closed.*	**Inspection.** Numbers only, no personal data about beneficiaries.	Must	3
US-8.08	**Issue a donation receipt** The system shall issue a PDF receipt with date, items, weight and NGO details for each NGO pickup. *As a business owner, I want a donation receipt for each NGO pickup so that I can document our contribution.*	**Test.** PDF receipt with date, items, weight and NGO details.	Should	2

4.9 E9 Impact Dashboard and Reporting

4.9.1 Description and Priority

Shows every user type the impact of what it saves, in kilograms, Egyptian pounds, CO_2e and meals, with a published method. Its design rests on three rules. First, **only verified actions count**: a pickup confirmed by code, a donation confirmed by the NGO, or a pantry item marked as used. Second, **every number has a published method**: each food category carries a weight, a CO_2e factor and an average price, stored in a versioned table that admins update from cited sources. Third, **each user type sees what helps them decide**, from a household's weekly waste rate to a ministry's district-level totals.

For each verified item i with weight w_i (kg) in food category $c(i)$:

$$\text{CO}_2\text{e avoided} = \sum_i w_i \times EF_{c(i)} \qquad \text{Money saved} = \sum_i (\text{original price}_i - \text{price paid}_i) \tag{4.1}$$

where EF_c is the emission factor (kg CO_2e per kg of food) for category c. Meals provided use a single documented conversion from kilograms, shown on the "How we calculate" screen.

The feature has 11 stories worth 49 points (5 Must, 5 Should, 1 Could). Built during the course: US-9.01 (Sprint 3), US-9.02 (Sprint 3), US-9.03 (Sprint 3), US-9.04 (Sprint 3).

Priority: High. Benefit 8, penalty 7, cost 5, risk 4 (1 = low, 9 = high).

4.9.2 Stimulus/Response Sequences

Related use cases: UC-17 to UC-19, UC-30, UC-41, UC-43, UC-50 (Appendix B).

#	Stimulus (user or system action)	Response (system behaviour)
1	A verified action is completed (pickup, donation or item used).	The system writes exactly one impact record using the current factor version.
2	The user opens the Impact tab.	The system shows totals for this week, this month and all time, split by Prevent, Sell and Donate.
3	The user taps *How we calculate*.	The system shows the factor, source and version date for each food category.
4	A business owner requests a CSR report.	The system generates a PDF with verified totals, method and period within 30 s.

4.9.3 Functional Requirements

ID	Requirement and user story	Acceptance test	Priority	Pts
US-9.01	**View personal impact** The system shall show each user's kg of food saved, EGP saved, CO_2e avoided and meals provided for this week, this month and all time. *As a user, I want a personal dashboard of kg of food saved, money saved (EGP), CO_2e avoided and meals provided so that I can see my contribution.*	**Test.** Totals for this week, month and all time; split by Prevent, Sell and Donate.	Must	5
US-9.02	**Explain the calculation** The system shall show the impact factor, its source and its version date for each food category. *As a user, I want a "How we calculate" screen so that I can trust the numbers.*	**Inspection.** Shows the factors per category, their source and version date.	Must	3
US-9.03	**Count only verified actions** The system shall record impact only for verified actions and count each action once. *As the platform, I want impact recorded only for verified actions and counted once so that totals are credible.*	**Test.** A bag counts once for the platform, even though both buyer and business see it; unconfirmed orders add nothing.	Must	5
US-9.04	**View the waste rate** The system shall show a monthly waste-rate trend against the Egyptian benchmark of 163 kg per person per year scaled to the household. *As a household user, I want my waste rate (food thrown away versus used) as a monthly trend, compared with the Egyptian average for my household size, so that I see whether I am improving.*	**Test.** Based on US-2.10, 2.08 data; benchmark of 163 kg per person per year scaled to my household.	Must	5

ID	Requirement and user story	Acceptance test	Priority	Pts
US-9.05	**Get a monthly insight [AI]** The system shall generate at most 3 monthly waste-reduction tips, each backed by the user's own pantry data. *As a household user, I want a monthly insight in plain language, such as "you throw away most bread on Thursdays; try buying 2 fewer loaves", so that I know what to change.*	**Analysis.** LLM summarises my own pantry data only; gives at most 3 concrete tips; no tip without supporting data.	Should	5
US-9.06	**Set a waste goal** The system shall track progress toward a user's waste-reduction goal and notify the user when it is reached or at risk. *As a household user, I want to set a goal, such as cutting waste by 20% this month, so that I stay motivated.*	**Test.** Progress bar; notification when the goal is reached or at risk.	Should	3
US-9.07	**View the business dashboard** The system shall show surplus listed, sold, donated and wasted, revenue recovered and sell-through rate by item, day and branch. *As a business manager, I want a dashboard of surplus listed, sold, donated and wasted, revenue recovered and sell-through rate, by item, day and branch, so that I can reduce over-production.*	**Demonstration.** Filters by date range and branch; charts update daily.	Must	5
US-9.08	**Export a CSR report** The system shall generate a PDF sustainability report with verified totals, method and period within 30 s. *As a business owner, I want to export a sustainability (CSR/ESG) report with verified figures so that I can share our results with management and customers.*	**Test.** PDF with totals, method and period; generated in under 30 s.	Should	5
US-9.09	**View the NGO dashboard** The system shall show each NGO's food received, meals distributed and donor reliability by month, exportable to CSV. *As an NGO coordinator, I want a dashboard of food received, meals distributed and donor reliability so that we can plan and report to our funders.*	**Demonstration.** Monthly view; export to CSV.	Should	3

ID	Requirement and user story	Acceptance test	Priority	Pts
US-9.10	**Export anonymised impact data** The system shall export monthly totals by district in the UNEP Food Waste Index sectors, hiding any group smaller than 10 users. *As a government analyst, I want anonymised totals by district and month, split into household, food service and retail, so that the data supports SDG 12.3 reporting.*	**Test.** Categories follow the UNEP Food Waste Index sectors; no group smaller than 10 users is shown; CSV export.	Should	8
US-9.11	**Share an impact card** The system shall create a shareable impact card that contains no personal data beyond the user's first name. *As a user, I want to share an impact card on social media so that I can encourage friends to join.*	**Inspection.** Image card in Arabic or English with no personal data beyond my first name.	Could	2

4.10 E10 Engagement, Ramadan and Events

4.10.1 Description and Priority

Helps the food-saving habit stick and covers the moments when waste peaks in Egypt: Ramadan, iftars, weddings and other gatherings. All stories are Should or Could; none is in the MVP.

The feature has 6 stories worth 25 points (0 Must, 3 Should, 3 Could). No story of this feature is built during the course.

Priority: Low. Benefit 4, penalty 2, cost 5, risk 3 (1 = low, 9 = high).

4.10.2 Stimulus/Response Sequences

Related use cases: UC-12, UC-19 (Appendix B).

#	Stimulus (user or system action)	Response (system behaviour)
1	The Hijri calendar reaches the first day of Ramadan.	The system turns on Ramadan mode with iftar and suhoor recipes and the portion planner.
2	A host enters the number of adult and child guests.	The system recommends quantities per dish type.
3	A referred friend completes a first verified action.	The system rewards both users.

4.10.3 Functional Requirements

ID	Requirement and user story	Acceptance test	Priority	Pts
US-10.01	**Earn badges and streaks** The system shall award badges and keep a weekly-check streak that survives one missed week per month. *As a household user, I want badges and streaks for completing my weekly pantry check and using food before it expires so that the habit sticks.*	**Test.** Streak shown on the home screen; streak survives one missed week per month.	Should	3
US-10.02	**Join a neighbourhood challenge** The system shall rank users who opt in by kg saved per person, showing display names only. *As a household user, I want opt-in neighbourhood challenges and a leaderboard so that saving food feels social.*	**Test.** Ranked by kg saved per person; only users who opt in appear; display names only.	Could	5
US-10.03	**Use Ramadan mode [AI]** The system shall switch Ramadan mode on and off by the Hijri calendar. *As a household user, I want a Ramadan mode with iftar and suhoor recipes, and a portion planner so that I waste less in the month when waste peaks.*	**Test.** Starts and ends by the Hijri calendar; the planner suggests quantities for the number of guests.	Should	8
US-10.04	**Plan portions for guests [AI]** The system shall recommend quantities per dish type for the number of adult and child guests. *As a host, I want to enter the number of guests for a gathering and get recommended quantities so that I cook enough without large leftovers.*	**Analysis.** Suggestions per dish type, adjusted for adults and children.	Should	5
US-10.05	**Read storage tips** The system shall show at most one team-reviewed storage tip per week. *As a household user, I want short tips on storage and labels in Egyptian Arabic so that I learn to keep food longer.*	**Inspection.** One tip per week at most; tips reviewed by the team before publishing.	Could	2
US-10.06	**Invite friends** The system shall reward both users after a referred friend's first verified action. *As a user, I want to invite friends with a referral link so that more people join.*	**Test.** Both users get a reward, such as a discount on the next bag, after the friend's first verified action.	Could	2

4.11 E11 Notifications

4.11.1 Description and Priority

Delivers the right alert to the right user at the right time, respecting preferences and quiet hours, and keeps a 30-day history in the app.

The feature has 5 stories worth 12 points (4 Must, 1 Should, 0 Could). Built during the course: US-11.01 (Sprint 1).

Priority: Medium. Benefit 6, penalty 6, cost 3, risk 2 (1 = low, 9 = high).

4.11.2 Stimulus/Response Sequences

Related use cases: UC-11, UC-21, UC-25 (Appendix B).

#	Stimulus (user or system action)	Response (system behaviour)
1	An event the user subscribed to occurs (digest, reservation, pickup).	The system sends a push notification that opens the related screen.
2	A notification is due during the user's quiet hours.	The system holds it until quiet hours end, unless it is a pickup reminder.
3	30 min before a pickup window ends and the order is not collected.	The system sends a pickup reminder.
4	The user opens the notification centre.	The system shows the last 30 days of notifications and clears the unread count.

4.11.3 Functional Requirements

ID	Requirement and user story	Acceptance test	Priority	Pts
US-11.01	**Receive push notifications** The system shall send push notifications for expiry digests, reservations and pickups that open the related screen. *As a user, I want push notifications for expiry digests, reservations and pickups so that I do not miss anything important.*	**Test.** Delivered through Firebase Cloud Messaging; deep-link to the right screen.	Must	3
US-11.02	**Set notification preferences** The system shall respect each user's notification choices and quiet hours (default 11 pm to 8 am), except for pickup reminders. *As a user, I want to choose which notifications I get and set quiet hours so that the app does not disturb me.*	**Test.** Toggle per type; default quiet hours 11 pm to 8 am, except pickup reminders.	Must	2
US-11.03	**Get deal alerts** The system shall send at most 3 deal alerts per day for a user's chosen stores, area and categories. *As a deal seeker, I want deal alerts for my favourite stores and area so that I catch good bags early.*	**Test.** Radius and categories configurable; at most 3 deal alerts per day.	Should	3

ID	Requirement and user story	Acceptance test	Priority	Pts
US-11.04	**Get a pickup reminder** The system shall remind a buyer 30 min before the pickup window ends if the order is not collected. *As a deal seeker, I want a reminder 30 min before my pickup window ends so that I do not miss it.*	**Test.** Sent only if the order is not yet collected.	Must	2
US-11.05	**View past notifications** The system shall keep each user's notifications for 30 days in an in-app notification centre. *As a user, I want an in-app notification centre so that I can see past alerts.*	**Demonstration.** Last 30 days kept; unread count on the icon.	Must	2

4.12 E12 Trust, Safety and Moderation

4.12.1 Description and Priority

Makes food and businesses on the platform trustworthy: automatic screening of listings, reporting, verified badges and mandatory allergens. The penalty of leaving it out is the highest of any feature, because one food-safety incident could stop the service (R-06).

The feature has 4 stories worth 12 points (4 Must, 0 Should, 0 Could). Built during the course: US-12.04 (Sprint 3).

Priority: High. Benefit 7, penalty 9, cost 4, risk 4 (1 = low, 9 = high).

4.12.2 Stimulus/Response Sequences

Related use cases: UC-31, UC-39, UC-44 (Appendix B).

#	Stimulus (user or system action)	Response (system behaviour)
1	Business staff submit a listing.	The system screens text and photo before publishing; flagged items go to human review instead of going live.
2	A user reports another user, post or listing.	The system records the report and later tells the reporter the outcome.
3	A user with a peanut allergy opens a listing that contains peanuts.	The system shows a clear allergy warning.

4.12.3 Functional Requirements

ID	Requirement and user story	Acceptance test	Priority	Pts
US-12.01	**Screen content automatically** The system shall check every listing against a list of prohibited items and words before publishing and hold flagged items for human review. *As the platform, I want every listing screened automatically for prohibited, unsafe or inappropriate content so that users can trust what they see.*	**Test.** Text is checked against the prohibited list (editable by administrators) before publishing; flagged items go to the human review queue (US-13.02) instead of going live.	Must	5
US-12.02	**Report a user or content** The system shall accept reports with a reason and tell the reporter the outcome. *As a user, I want to report a user or listing so that problems are reviewed.*	**Test.** Report reasons list; the reporter is told the outcome.	Must	3
US-12.03	**Show a verified badge** The system shall show a verified badge only on businesses approved by an administrator. *As a deal seeker, I want a "verified" badge on approved businesses so that I know they are licensed.*	**Test.** Badge shown only after US-13.01 approval.	Must	2
US-12.04	**Declare allergens** The system shall require allergens on every listing and warn users whose profile allergies match. *As a user with allergies, I want allergens declared and clearly shown on every listing so that I stay safe.*	**Test.** Allergen field required; warning if it matches allergies in my profile.	Must	2

4.13 E13 Administration

4.13.1 Description and Priority

Gives the Elha2 operations team the tools to verify organisations, moderate content, resolve disputes, configure rules without new code, and manage the recipe database and food catalogue.

The feature has 6 stories worth 22 points (4 Must, 2 Should, 0 Could). Built during the course: US-13.01 (Sprint 1).

Priority: Medium. Benefit 5, penalty 7, cost 4, risk 2 (1 = low, 9 = high).

4.13.2 Stimulus/Response Sequences

Related use cases: UC-44 to UC-49 (Appendix B).

#	Stimulus (user or system action)	Response (system behaviour)
1	An administrator opens a pending business registration.	The system shows the documents with approve, reject-with-reason and request-information actions.
2	The administrator approves the business.	The system marks it verified, shows the badge and lets its branches list surplus.
3	The administrator changes a platform setting.	The system saves a new version of the setting with the administrator's name and the time.
4	The administrator resolves a dispute with a partial refund.	The system issues the refund through the payment provider and logs the decision.

4.13.3 Functional Requirements

ID	Requirement and user story	Acceptance test	Priority	Pts
US-13.01	**Verify organisations** The system shall let an administrator approve, reject with a reason, or query each business and NGO registration. *As an administrator, I want to review and approve business and NGO registrations so that only legitimate organisations join.*	**Demonstration.** View documents; approve, reject with reason, or request more information.	Must	3
US-13.02	**Moderate flagged content** The system shall list flagged and reported content by severity with approve, remove, warn and suspend actions. *As an administrator, I want a moderation queue of flagged and reported content so that I can act quickly.*	**Demonstration.** Sorted by severity; actions: approve, remove, warn or suspend user.	Must	3
US-13.03	**Resolve disputes** The system shall let an administrator issue a partial or full refund from a dispute and log the decision. *As an administrator, I want to handle disputes and issue refunds so that customers and businesses are treated fairly.*	**Test.** Full history of the order and chat; partial or full refund; decision logged.	Must	5
US-13.04	**Configure settings** The system shall let administrators change commission, discount limits, cancellation rules, impact factors and AI thresholds, versioning and logging every change. *As an administrator, I want to configure platform settings (commission, discount limits, cancellation rules, impact factors, AI confidence thresholds) so that rules change without new code.*	**Test.** Every change is versioned and logged with who made it.	Must	3

ID	Requirement and user story	Acceptance test	Priority	Pts
US-13.05	**View platform analytics** The system shall report active users, orders, sales value, retention and kg saved by day, week and month, exportable to CSV. *As an administrator, I want platform analytics (active users, orders, sales value, retention, kg saved) so that we can track the business.*	**Demonstration.** Daily, weekly and monthly views; export to CSV.	Should	5
US-13.06	**Manage catalogue and recipes** The system shall let administrators add, edit and retire recipes and food catalogue entries in Arabic and English. *As an administrator, I want to manage the recipe database and the food catalogue with shelf-life data so that suggestions and estimates stay accurate.*	**Demonstration.** Add, edit and retire entries in Arabic and English.	Should	3

4.14 E14 AI Platform (Enablers)

4.14.1 Description and Priority

Enabler stories, written from the development team's point of view. They make the AI features accurate, private and affordable: one AI gateway for every model call, a labelled Egyptian test set, personal-data masking, correction logging and cost caps.

The feature has 5 stories worth 22 points (4 Must, 1 Should, 0 Could). Built during the course: US-14.01 (Sprint 1), US-14.02 (Sprint 2), US-14.03 (Sprint 1).

Priority: High. Benefit 7, penalty 8, cost 6, risk 7 (1 = low, 9 = high).

4.14.2 Stimulus/Response Sequences

Related use cases: UC-04 and every use case with AI services as an actor (Appendix B).

#	Stimulus (user or system action)	Response (system behaviour)
1	Any feature requests an AI result.	The AI gateway applies the versioned prompt, a 15 s time-out and schema validation, and returns a result with a confidence and a fallback flag.
2	The AI provider fails or times out.	The gateway returns the non-AI fallback (manual entry or a rule-based result) and logs the failure.
3	A release build runs in CI.	The pipeline measures accuracy on the labelled test set and publishes the report.
4	AI spending reaches 80% of the monthly budget.	The system alerts the team and enables stricter caching.

4.14.3 Functional Requirements

ID	Requirement and user story	Acceptance test	Priority	Pts
US-14.01	**Route all AI calls through a gateway** The system shall send every AI request through one gateway that applies a 15 s time-out and a non-AI fallback. *As a developer, I want one AI gateway service that wraps all LLM and vision providers so that we can change provider, version prompts and fall back safely.*	**Test.** Prompt templates versioned in the repository; time-out of 15 s with a fallback (manual entry or rule-based result); provider set in configuration.	Must	8
US-14.02	**Measure AI accuracy** The system shall produce an AI accuracy report from a labelled set of at least 300 receipts, 200 photos and 100 voice notes on every release. *As the team, I want a labelled test set so that AI accuracy is measured at every release.*	**Analysis.** Report produced automatically in the CI pipeline.	Must	5
US-14.03	**Mask personal data** The system shall remove names, phone numbers and card digits from receipt text on the phone before any upload. *As a privacy-conscious user, I want personal details (names, phone numbers, card digits) removed from receipts before they are sent to a cloud AI service so that my data is protected.*	**Test.** Masking runs on the phone before upload; verified on the test set.	Must	3
US-14.04	**Log AI corrections** The system shall log user corrections to AI results, with consent and without personal data. *As the team, I want user corrections to AI results logged, with consent, so that we can improve prompts and models.*	**Inspection.** Corrections stored without personal data; reviewed every sprint.	Should	3
US-14.05	**Cap AI cost** The system shall cap AI spending per active user per month and alert the team at 80% of the budget. *As the team, I want AI costs monitored and limited so that the service stays affordable.*	**Test.** Monthly budget per active user; repeated results (e.g. the same barcode or receipt line) cached; alert at 80% of budget.	Must	3

5. Other Nonfunctional Requirements

Non-functional requirements use the same form as the functional ones: an identifier, a "shall" statement with a measurable target, a verification method and a MoSCoW priority. Each target cites the benchmark, standard or law it is based on.

5.1 Performance Requirements

Response-time targets follow the classic limits for keeping a user's attention (about 1 s for an uninterrupted flow, 10 s before attention is lost) [11] and evidence that mobile users abandon slow screens [12].

ID	Requirement	Verification	Priority
NFR-01	The mobile app shall start from cold in 3 s or less on an Android phone with 3 GB of RAM. Android vitals flags cold starts of 5 s or more as excessive [13].	Test	Must
NFR-02	The backend API shall answer 95% of read requests within 500 ms and 95% of write requests within 1 s under the expected pilot load [11].	Test	Must
NFR-03	The system shall return nearby-listing searches within a 20 km radius in 2 s or less [11, 12].	Test	Must
NFR-04	The system shall show the review screen within 10 s of a receipt photo on a 4G connection, the limit for keeping a user's attention on a task [11].	Test	Must
NFR-05	The system shall support 50,000 registered and 2,000 concurrent users with a stateless backend that scales horizontally to 10 times that load through the platform's automatic instance scaling [31].	Analysis	Should
NFR-06	The system shall reserve each bag at most once, with zero overselling in a test of 100 simultaneous reservations on one listing, using transactional updates [14].	Test	Must

5.2 Safety Requirements

The main hazards are food poisoning, allergic reactions and unsafe AI advice. These requirements complement the listing screening and allergen warnings of feature E12 (Section 4.12), the date-label explanations in US-3.04 and risk R-06.

ID	Requirement	Verification	Priority
NFR-07	The system shall filter recipe and chat output for unsafe food handling, following the WHO Five Keys to Safer Food [15], and shall not give medical or nutritional advice.	Test	Must

ID	Requirement	Verification	Priority
NFR-08	The system shall allow sales only by businesses with a valid food-safety licence and require allergen declarations, as required by the National Food Safety Authority [16].	Inspection	Must

5.3 Security Requirements

Users are identified by phone number and a one-time code (US-1.01); there are no passwords. Each user acts within one or more roles (household user, business owner, manager or counter staff, NGO coordinator, administrator, government analyst).

ID	Requirement	Verification	Priority
NFR-09	The system shall encrypt all network traffic with TLS 1.2 or higher [17] and encrypt the database and file storage at rest.	Inspection	Must
NFR-10	The system shall restrict each user to the data and actions of their role, and staff to their own business or NGO, to prevent broken access control, the top web application risk [8].	Test	Must
NFR-11	The system shall process card payments only through Paymob's hosted or tokenised flow and store no card numbers [9].	Inspection	Must
NFR-12	The system shall rate-limit login, one-time code and API requests, and lock an account for 1 h after 5 failed code attempts [18].	Test	Must
NFR-13	The system shall obtain consent for, limit the purpose of, and provide access to and deletion of personal data [6].	Inspection	Must
NFR-14	The system shall delete photos sent for AI processing within 24 h unless the user opts in to keep them, so that personal data is not kept longer than needed [6].	Test	Must
NFR-15	The system shall publish only aggregated data in which every group contains at least 10 users, so that no household can be singled out [19].	Test	Must

5.4 Software Quality Attributes

When quality attributes conflict, we rank them as follows: **safety and privacy** first, then **usability for first-time household users**, then **reliability** of reservations and reminders, then **performance**, then **maintainability** and **portability**. For AI features, a correct result confirmed by the user is preferred over a fully automatic one.

5.4.1 Usability

ID	Requirement	Verification	Priority
NFR-16	The system shall use Arabic by default, with a full right-to-left layout [10] and Egyptian colloquial wording on household screens.	Inspection	Must
NFR-17	The system shall meet WCAG 2.1 level AA, including text scaling to 200% and screen-reader labels on all controls [7].	Inspection	Should
NFR-18	The system shall show a text label with the icon of every main action [20] and offer voice input for pantry logging.	Inspection	Could

5.4.2 Reliability

ID	Requirement	Verification	Priority
NFR-19	The system shall run expiry reminders and NGO offers within 5 min of their scheduled time and retry a failed job 3 times before alerting the team [21].	Test	Must
NFR-20	The system shall back up all data daily with encryption and restore service within 4 h (recovery time objective) with at most 24 h of data loss (recovery point objective) [22].	Demonstration	Must
NFR-21	The mobile app shall let users view the pantry, scan barcodes and add items while offline, and sync those changes without loss on reconnection [23].	Test	Must
NFR-22	The system shall complete every AI-supported task through a manual or rule-based path when the AI service fails or times out [24].	Test	Must

5.4.3 Availability

ID	Requirement	Verification	Priority
NFR-23	The system shall be available 99.5% of each month, excluding maintenance announced 24 h ahead and scheduled outside 6 pm to midnight. The target sits below the 99.95% monthly uptime of the planned hosting platform [25].	Analysis	Must

5.4.4 Maintainability and Testability

ID	Requirement	Verification	Priority
NFR-24	The backend shall keep unit-test coverage at 70% or more, between Google's "acceptable" (60%) and "commendable" (75%) levels [26], with automated tests and a code review on every merge.	Analysis	Should
NFR-25	The system shall publish an OpenAPI specification generated from the code at every release [27].	Inspection	Should

ID	Requirement	Verification	Priority
NFR-26	The system shall log errors centrally and alert the team when the error rate exceeds 2% for 5 min [21].	Test	Should

5.4.5 Portability

ID	Requirement	Verification	Priority
NFR-27	The mobile app shall run on Android 8.0+ and iOS 15+, and the web portal on the latest 2 versions of Chrome, Edge, Safari and Firefox, covering the devices in use in Egypt [28].	Test	Must
NFR-28	The mobile app shall install in 60 MB or less, upload images of 300 KB or less, and use no background location tracking.	Inspection	Should

5.4.6 AI Quality and Responsible AI

ID	Requirement	Verification	Priority
NFR-29	The AI features shall read at least 90% of receipt lines, find at least 80% of items in fridge photos and extract at least 85% of voice-note items correctly on the labelled test set (US-14.02). The receipt target is in line with results on the SROIE receipt benchmark [29].	Analysis	Must
NFR-30	The system shall show a confidence level on every AI result, let the user edit it, and show other users nothing AI-generated before a person confirms it [24].	Demonstration	Must

5.5 Business Rules

ID	Rule	Enforced by
BR-01	Only a business or NGO verified by an administrator may list surplus or receive donation offers.	US-1.06, 1.07, 12.03, 13.01
BR-02	A marketplace listing must be discounted by 30% to 80% of its original value (configurable).	US-6.01
BR-03	Every listing must declare its allergens.	US-12.04; NFR-08
BR-04	Counter staff may only view the day's orders and confirm pickups; only owners and managers may create listings, change prices or see payouts.	US-6.03; NFR-10
BR-05	A buyer gets a full refund if a reservation is cancelled at least 1 h before the pickup window (configurable), and always when the business cancels.	US-7.04, 7.05
BR-06	Three no-shows within 30 days block a user from reserving for 7 days.	US-7.04

ID	Rule	Enforced by
BR-07	Cash at pickup is allowed only after 2 completed pickups and no no-show in the last 30 days.	US-7.02
BR-08	An NGO has 30 min to accept a donation offer before it passes to the next matched NGO.	US-8.02
BR-09	Impact is recorded only for verified actions, and each action is counted once for the whole platform.	US-9.03
BR-10	Only buyers with a confirmed pickup may rate a store.	US-5.08
BR-11	Elha2 keeps a commission on each marketplace sale; donations are free of charge.	US-7.07
BR-12	Government analysts see only aggregates in which every group has at least 10 users.	US-9.10; NFR-15

6. Other Requirements

This chapter holds the requirements that do not fit the earlier chapters (database, localisation and legal requirements), followed by the sections that Milestone 1 of the course asks for: the global and societal considerations, the identified risks, the features we will fully implement during the course, and the conclusion. The template allows project-specific sections here.

6.1 Logical Database Requirements

Data is stored in PostgreSQL, hosted on Supabase, with the PostGIS extension for locations. The main entities are listed below. Their attributes, types and relationships are designed in the SDD (Chapter 4, Data Design, with a UML class diagram).

Entity	Key attributes	Integrity rules
User	ID, phone, name, language, area, roles, allergies	Phone number unique; personal data deletable (US-1.08)
Household	ID, members, size, dietary preferences	A user belongs to at most one household
PantryItem	ID, household, catalogue item, quantity, unit, storage place, expiry date, expiry source (printed or estimated), status, waste reason	Status is one of in stock, used, donated, thrown away; every change is kept for impact
FoodCatalogueItem	ID, Arabic and English names, category, barcode, shelf life per storage place	Managed by administrators (US-13.06)
ImpactFactor	Category, kg CO_2e per kg, average price, source, version, valid from	Versioned and never overwritten (US-9.02)
Business, Branch	ID, documents, verification status, location, opening hours, donation partner	A branch cannot list until its business is verified
Listing	ID, branch, type, price, original value, quantity, pickup window, allergens, photo, status	Quantity never below zero; allergens required
Order	ID, user, listing, quantity, amount, payment status, pickup code, status, verified at	Pickup code unique per order
Payment	ID, order, provider, provider reference, amount, status	No card data stored
NGO	ID, registration number, service area, accepted food types, capacity, hours	Must be verified before receiving offers
DonationOffer	ID, listing or pantry item, NGO, offered at, response, pickup time, meals distributed	One active offer per item at a time
ImpactRecord	ID, source action, user, business, NGO, kg, EGP, CO_2e, factor version, verified at	Written once per verified action (US-9.03)
Notification	ID, user, type, content, sent at, read at	Kept for 30 days
AIRequestLog	ID, feature, model version, latency, cost, confidence, user correction	Holds no personal data

Data is kept while an account is active. Personal data is deleted within 30 days of a deletion request, AI photos within 24 hours (NFR-14), and impact records are kept only in anonymised form.

6.2 Internationalisation Requirements

Arabic is the primary language and English the secondary one. All user-facing text lives in translation files, never in code, so that further languages or markets can be added through configuration (Section 6.4, Global).

ID	Requirement	Verification	Priority
NFR-31	The system shall show prices in EGP, times in the Africa/Cairo time zone, and dates in both the Gregorian and Hijri calendars, using standard locale data [30].	Inspection	Must

6.3 Legal and Regulatory Requirements

Law or standard	Obligation for Elha2	Requirements
Egypt Personal Data Protection Law No. 151 of 2020 [6]	Consent, purpose limitation, access and deletion of personal data; breach notification	NFR-13 to NFR-15; US-1.08, 14.03
NFSA food-safety rules [16]	Only licensed businesses sell food; allergens declared	NFR-08; US-1.06, 12.04
PCI DSS [9]	No card data stored or processed by Elha2	NFR-11; US-7.01
WCAG 2.1 [7]	Accessible user interfaces	NFR-17, NFR-18
Consumer protection and terms of use	Clear prices, refund rules and terms accepted at sign-up	BR-05; US-7.04, 7.05

6.4 Global, Cultural, Social, Environmental, Economic, Health, Safety and Welfare Considerations

Elha2's requirements address eight areas of impact. Table 6.1 traces each area to the requirements that address it, so each concern can be designed, built and tested like any other requirement.

Table 6.1: Requirements by area of impact.

Area	How Elha2 addresses it	Requirements
Global	Targets SDG 12.3 (halve food waste by 2030) and supports SDG 2 (zero hunger) and SDG 13 (climate action). Government exports use the UNEP Food Waste Index sectors (household, food service, retail), so Egyptian data can be compared with other countries. Language, currency, time zone and calendar are configuration, not code, so the product can be localised for other markets.	US-9.10; NFR-15, NFR-31; US-1.02
Cultural	Arabic is the default language, with right-to-left layout and Egyptian colloquial wording. Voice logging understands Egyptian Arabic. Recipes start from Egyptian dishes, such as turning stale bread into fattah. Ramadan mode follows the Hijri calendar, with iftar and suhoor recipes and a portion planner for guests, when waste peaks. Cash at pickup matches local payment habits.	NFR-16, NFR-31; US-1.02, 2.06, 4.01, 4.07, 7.02, 10.03, 10.04
Social	NGOs record meals as numbers only, protecting the dignity and privacy of beneficiaries. Icons, voice input and accessibility support include users with lower literacy or disabilities. Discounted surplus makes good food affordable for students and low-income workers.	E8; US-8.07; NFR-17, NFR-18; US-5.01, 5.04
Environmental	Every verified action is converted into CO_2e avoided, using published, versioned factors, so users see the climate effect of what they save. Waste-rate feedback and goals help households change habits. Pickup-only operation avoids a delivery fleet, and the app avoids background GPS to save battery.	US-9.01 to US-9.06; NFR-28
Economic	Households see the money they save in EGP. Businesses recover revenue from unsold stock and cut disposal costs, and get CSR reports and donation receipts. A commission on marketplace sales funds the platform. The stack uses free tiers, and AI cost is capped per active user, so the service stays affordable to run.	US-9.01, 9.07, 9.08, 8.08, 6.01, 7.07, 14.05; NFR-05
Public health	Clear "use by" versus "best before" labels reduce both waste and food poisoning. Allergens are mandatory and checked against the user's profile, and only licensed, verified businesses can sell. Leftover advice includes reheating safety notes, and the AI never gives medical or nutritional advice.	US-3.02, 3.04, 4.02, 12.04, 1.06, 13.01; NFR-07, NFR-08
Safety	Users can rate stores and report problems, and listings are screened before they go live. Every AI output is confirmed by a person and has a non-AI fallback.	US-5.08, 5.09, 12.01, 12.02; NFR-22, NFR-30
Public welfare	Personal data is handled under Egypt's PDPL, and users can delete their accounts. Data shared with government partners is aggregated and anonymised, giving policymakers evidence without exposing anyone. The impact method is public, disputes are handled fairly, and security controls protect all users.	NFR-09 to NFR-13, NFR-15; US-1.08, 9.02, 9.10, 13.03

6.4.1 Trade-offs and Possible Negative Effects

- **Impact data versus privacy.** Richer data would make national statistics more precise, but could identify households. We publish only aggregates of at least 10 users (NFR-15).
- **Automation versus accuracy.** AI removes typing but can be wrong. Every AI result is reviewed by the user before it is saved, and the user can always enter items by hand (NFR-22, NFR-30).
- **Digital divide.** Elha2 needs a smartphone, which excludes some of the people most affected by food insecurity. NGO routing reaches them through organisations rather than through the app.
- **Discounts versus donations.** Selling surplus could reduce what reaches charities. Auto-donation sends whatever is not sold or collected to NGOs (US-8.02).

6.5 Risks

We identified 13 risks across technology, AI, external partners, safety and the business model. The most serious is AI accuracy on Egyptian receipts (R-01). Likelihood and impact are rated Low, Medium or High; Figure 6.1 places each risk on the matrix, and Table 6.2 gives the mitigation and the fallback for each one. The register is reviewed at every sprint retrospective.

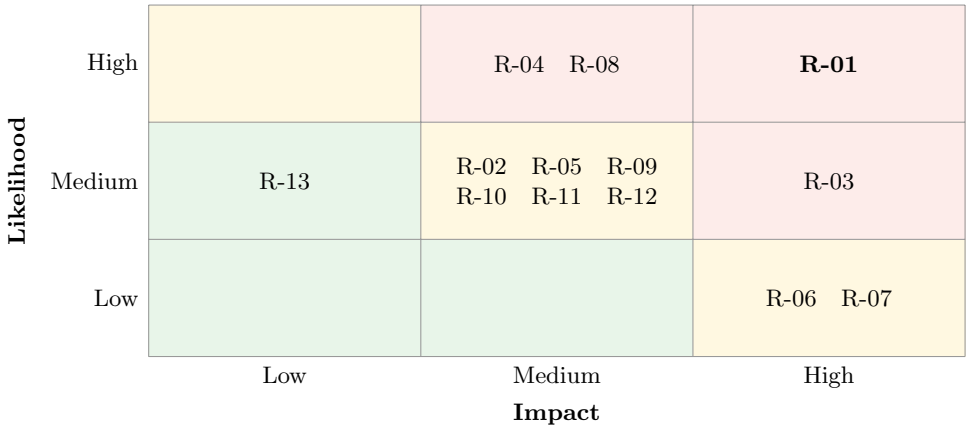


Figure 6.1: Risk matrix. Red cells need active mitigation from Sprint 1; amber cells are watched every sprint.

Table 6.2: Risk register (L = likelihood, I = impact; L = Low, M = Medium, H = High).

ID	Risk	Category	L	I	Mitigation and fallback
R-01	AI misreads Egyptian receipts (mixed Arabic and English, shop abbreviations, faded thermal paper) and misses the 90% accuracy target.	AI	H	H	Build the labelled test set in Sprint 1 (US-14.02); choose the LLM and prompts by measured accuracy; every result is reviewed by the user (US-2.02). *Fallback:* manual entry with autocomplete (US-2.07).
R-02	AI API cost or rate limits exceed our student budget.	Cost	M	M	OCR runs on the phone, so only short text is sent; cache repeated lines; cap cost per user (US-14.05); use a smaller model where accuracy allows. *Fallback:* limit scans per user per day during the demo.
R-03	Households stop keeping their pantry up to date.	Product	M	H	Automatic inputs, automatic deductions when food is cooked or donated, and a 30-second weekly check (E2). We measure check completion in usability tests. *Fallback:* items past their expected use are archived automatically.
R-04	We cannot activate a live payment account during the course (Paymob requires a registered company).	External	H	M	Cash at pickup in the course scope (US-7.02); in-app payment (US-7.01) after the course.
R-05	No real NGO or business joins the pilot in time for the demo.	External	M	M	Seed realistic demo accounts. *Fallback:* demonstrate with simulated partners.
R-06	Someone falls ill from surplus or donated food.	Safety, legal	L	H	Only verified, licensed businesses can sell; mandatory allergens; reporting; clear terms of use. *Fallback:* suspend the business involved.
R-07	A privacy breach exposes personal data or locations.	Security	L	H	Encryption, role-based access, masking of personal details before AI calls (NFR-09, NFR-10, NFR-14; US-14.03). *Fallback:* incident response, revoke tokens, notify users as PDPL requires.
R-08	Cold start: few listings attract few buyers, and few buyers attract few businesses.	Business	H	M	The household pantry gives value without any other users; launch in one area first (New Cairo, near campus). *Fallback:* partner-seeded listings.
R-09	A third-party service fails or changes its API (Supabase, Firebase, Maps, LLM provider).	Technical	M	M	All AI calls go through one gateway, so the provider can be swapped (US-14.01); non-AI fallbacks (NFR-22); pinned SDK versions.

ID	Risk	Category	L	I	Mitigation and fallback
R-10	Abuse: no-shows, fake listings or reselling donated food.	Trust	M	M	Organisation verification, pickup codes, no-show limits, ratings and reports, verified NGOs only.
R-11	Impact figures are challenged as inaccurate ("greenwashing").	Credibility	M	M	Count only verified actions, once (US-9.03); publish versioned factors with sources (US-9.02); use conservative estimates.
R-12	AI gives unsafe or culturally inappropriate output, such as poor reheating advice.	AI, safety	M	M	Curated Egyptian recipes first; output filters (NFR-07); human confirmation before anything AI-made is shown to others (NFR-30).
R-13	Offline edits conflict in a shared household pantry.	Technical	M	L	Per-field last-write-wins with a change log and an undo (US-2.08).

6.6 Features Implemented During the Course

During the course we will fully implement 39 user stories. This subset of the backlog covers one complete loop of the product: food enters the pantry with AI, is used before it expires, is sold or donated, and the verified result appears on an impact dashboard. The remaining 61 stories stay in the backlog for after the course.

6.6.1 Selection Criteria

- **One working end-to-end loop** across all three pillars (Prevent, Sell, Donate) and the impact dashboard, rather than a deep build of a single pillar.
- **Real AI in the core,** not mock-ups: on-device OCR, an LLM through the AI gateway, the shelf-life estimator and LLM-backed recipe suggestions, measured on our own test set.
- **Safety and privacy before reach.** Verification, allergens and masking of personal data are in the core, because the app cannot be tested with real users without them.

6.6.2 Sprint Plan

Sprint	Ends at	Goal	User stories	Points
1	Milestone 2	Foundations, manual pantry and AI receipt capture	US-1.01, 1.02, 1.04, 1.06, 1.07, 13.01, 2.07, 2.10, 2.11, 14.01, 11.01, 2.01, 2.02, 14.03	52
2	Milestone 3	Expiry, recipes and the marketplace	US-14.02, 3.01, 3.03, 3.04, 2.08, 2.09, 4.01, 4.05, 6.01, 5.01, 5.03, 5.04, 5.07, 7.02, 7.03	55
3	Milestone 4 (December 1)	Pickup, donation, safety and impact	US-6.04, 6.07, 8.01, 8.02, 8.04, 12.04, 9.01, 9.02, 9.03, 9.04	39
			39 stories	**146**

6.7 Conclusion

Milestone 1 turns the Elha2 idea into a scoped and specified system. This SRS defines the product, its six user types and its motivation. Its backlog contains 100 user stories in 14 features, 31 non-functional requirements and 12 business rules, modelled as 50 use cases in five UML use case diagrams (Appendix B). The companion Software Design Document [3] gives the component-based architecture that implements them.

This SRS and the SDD are living documents: we will update both at the end of every sprint and record each change in their revision histories.

Appendix A: Glossary

Terms, acronyms and abbreviations used in this SRS, in alphabetical order. Terms used only in the design are defined in the SDD.

Term	Definition
Acceptance test	The condition a requirement must meet to be accepted, with its verification method.
AI gateway	The single backend component through which every call to a hosted AI model passes (US-14.01).
Best before	Date until which food keeps its best quality; it is usually still safe after this date.
CO_2e	Carbon dioxide equivalent: greenhouse gas emissions expressed as the amount of CO_2 with the same warming effect.
Definition of Done	The checklist every story must pass before it counts as done: code reviewed and merged, tests and acceptance test passing, Arabic and English text in place, and correct right-to-left layout.
EGP	Egyptian pound.
Epic	A large group of related user stories; in this SRS each epic is one system feature (E1 to E14).
FCM	Firebase Cloud Messaging, Google's push notification service.
Kanban board	A board with To do, In progress and Done columns that shows the state of every story in a sprint.
LLM	Large language model.
MoSCoW	Prioritisation into Must, Should, Could and Won't (now).
MVP	Minimum viable product: the first release, containing all Must requirements.
NFSA	National Food Safety Authority of Egypt.
NGO	Non-governmental organisation; here, food banks and charities.
OCR	Optical character recognition.
OTP	One-time password sent by SMS.
PDPL	Egypt's Personal Data Protection Law No. 151 of 2020.
Pickup window	Time range in which a reserved item must be collected.
PostGIS	An extension of PostgreSQL that stores locations and answers distance queries.
Product backlog	The ordered list of all requirements for the product, written as user stories (Chapter 4).
Term	**Definition**
RTL	Right-to-left text layout, used for Arabic.
Scrum	An agile process in which a team delivers working software in short, fixed-length sprints.
SDG	United Nations Sustainable Development Goal.
Sprint	A fixed period of plan, build, test and review; the course has three, ending at Milestones 2, 3 and 4.
Sprint backlog	The stories chosen from the product backlog for one sprint.
Story point	Unit of relative effort used to estimate a user story.
STT	Speech-to-text.
Surprise bag	A bag of unsold food of a stated value whose exact contents are not fixed in advance.

Term	Definition
TBD	To be determined: an open decision, tracked in Appendix C.
Use by	Date after which food may be unsafe to eat.
Use case	A goal an actor achieves with the system, shown in a UML use case diagram (Appendix B).
User story	A requirement written as "As a [user], I want [goal] so that [benefit]".
Verified action	A food-saving action confirmed by a pickup code or by an NGO, or a pantry item marked as used; only these count toward impact.
VLM	Vision-language model: an AI model that understands both images and text.

Appendix B: Analysis Models

This appendix holds the analysis models of Elha2. The context diagram (Figure 2.1) and the automated pantry pipeline (Figure 2.2) are in Section 2.1. The design models (UML component, class and sequence diagrams) are in the SDD [3].

B.1 Use Case Model

The use case model shows who interacts with Elha2 and for what purpose. It is drawn as five UML use case diagrams, one per subsystem, with 50 use cases in total. Table B.1 traces every use case to the user stories in Chapter 4.

B.2 Actors

Actor	Type	Role in the system
App user	Primary	Any registered user; generalises every human role below
Household user	Primary	Manages the household pantry, cooks, donates and tracks waste
Deal seeker	Primary	Finds, reserves and collects discounted surplus
Business staff, Business owner	Primary	List surplus, hand over orders, manage the business account and reports
NGO coordinator	Primary	Receives donation offers, books pickups and reports meals distributed
Administrator	Primary	Verifies organisations, moderates content, resolves disputes, configures the platform
Government analyst	Primary	Exports anonymised, aggregated impact data
Scheduler	System (internal timer)	Starts time-based use cases: expiry reminders and routing of unsold food
AI services	System (external)	LLM, vision and speech models reached through the AI gateway
Payment provider	System (external)	Paymob: online payments and refunds
Product database	System (external)	Open Food Facts: product details for barcodes
Grocery order email	System (external)	Forwarded online grocery order confirmations
Authentication service	System (external)	Supabase Auth: phone number and one-time code

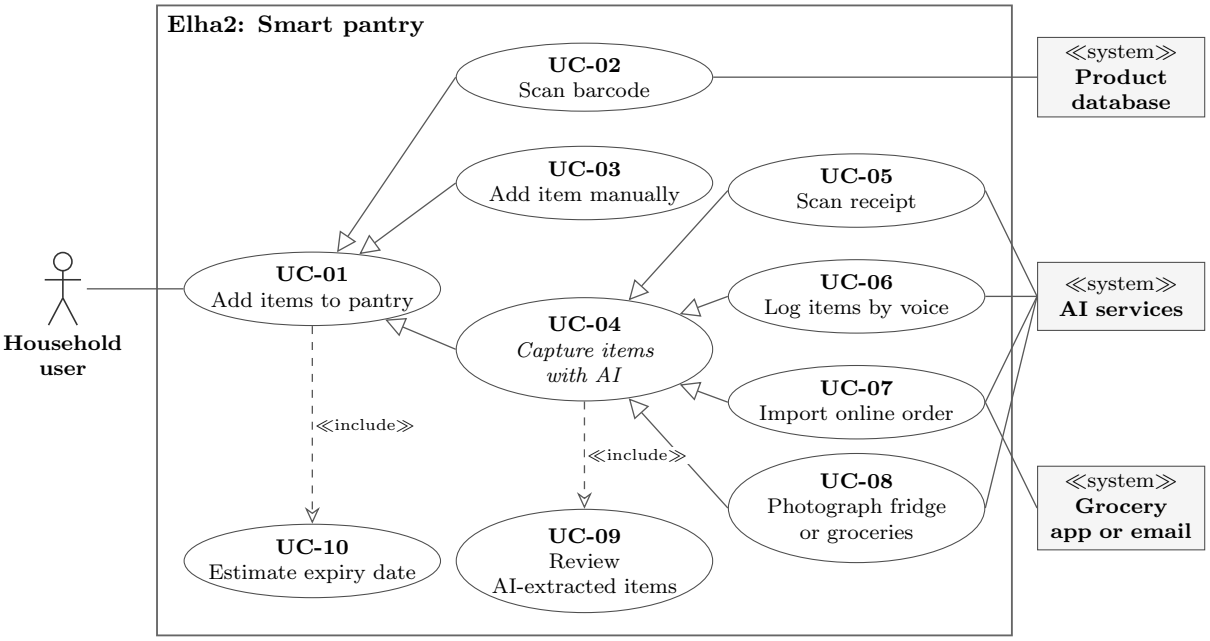


Figure B.1: Use case diagram 1: adding food to the smart pantry. All automatic inputs specialise UC-04, so every one of them includes the one-tap review (UC-09).

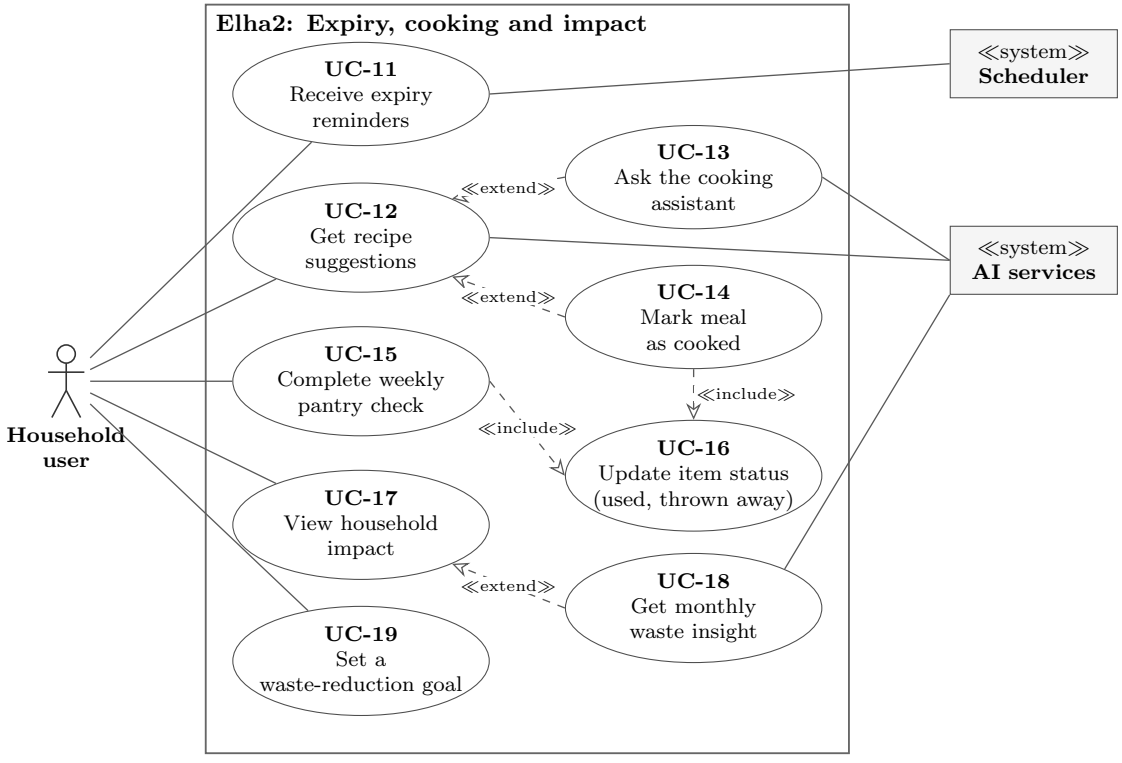


Figure B.2: Use case diagram 2: using food before it expires, and tracking household impact.

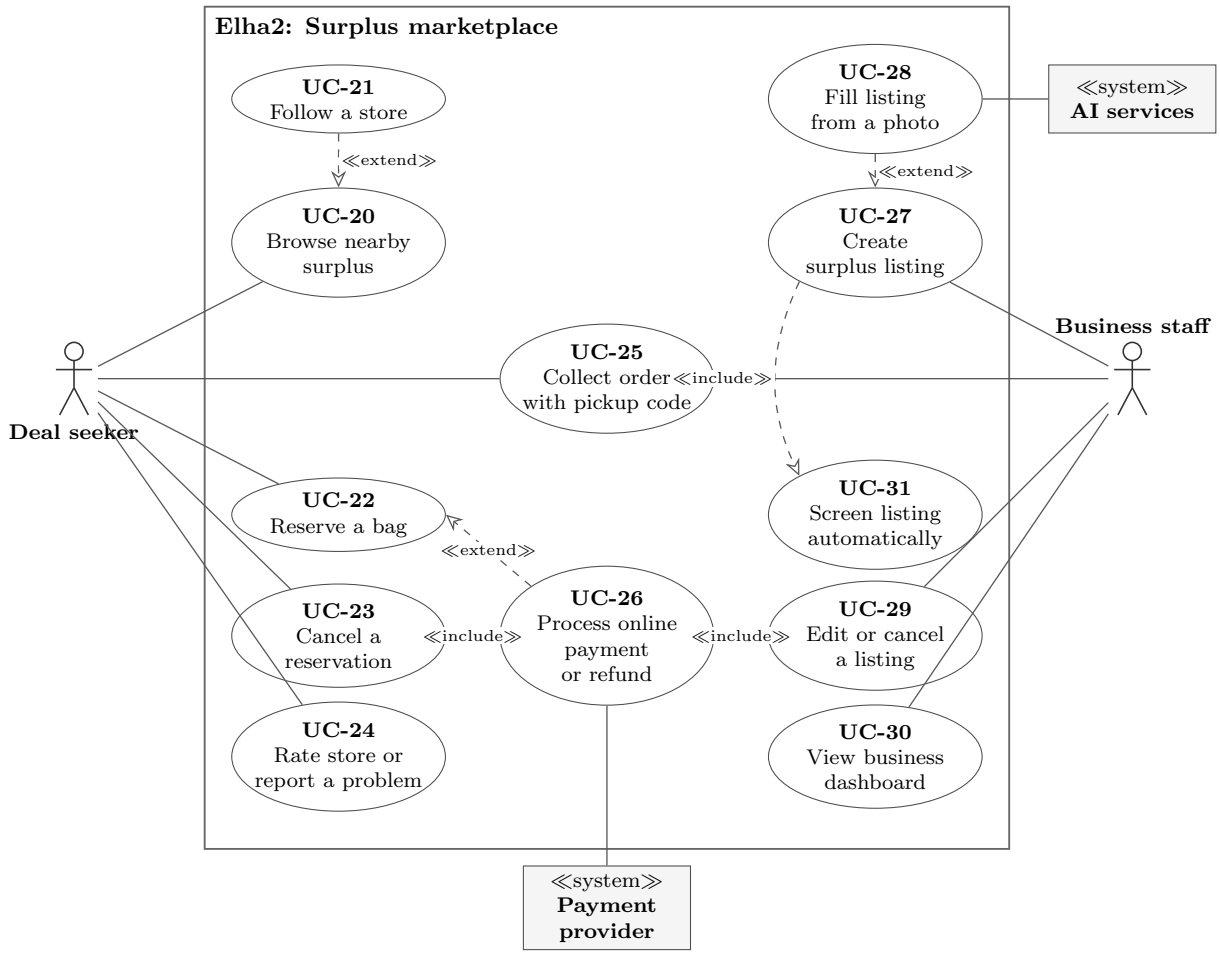


Figure B.3: Use case diagram 3: the surplus marketplace. Both the deal seeker and business staff take part in the pickup handover (UC-25), which is what makes an order count as verified impact.

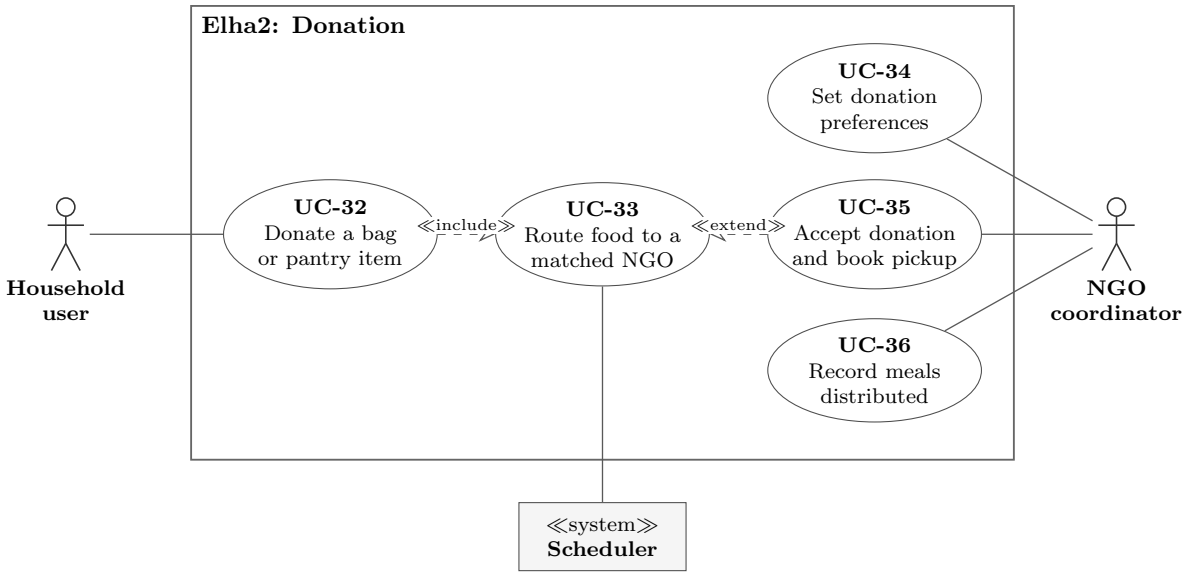


Figure B.4: Use case diagram 4: NGO donation. Unsold surplus and household donations both reach an NGO through UC-33.

B.3 Use Case Diagrams

B.4 Use Case Catalogue

Table B.1: Use case catalogue and traceability to user stories.

ID	Use case	Primary actor	Secondary	Stories
UC-01	Add items to pantry (generalises UC-02 to UC-08)	Household user	–	US-2.01–2.07
UC-02	Scan barcode	Household user	Product database	US-2.03
UC-03	Add item manually	Household user	–	US-2.07
UC-04	*Capture items with AI* (abstract)	Household user	AI services	US-14.01, 14.02
UC-05	Scan receipt	Household user	AI services	US-2.01, 14.03
UC-06	Log items by voice	Household user	AI services	US-2.06
UC-07	Import online order	Household user	AI services, grocery app	US-2.04
UC-08	Photograph fridge or groceries	Household user	AI services	US-2.05
UC-09	Review AI-extracted items	Household user	–	US-2.02
UC-10	Estimate expiry date	Household user	–	US-3.01, 3.02
UC-11	Receive expiry reminders	Scheduler	Household user	US-3.03, 3.04, 3.05, 3.06, 10.05, 11.01, 11.05

ID	Use case	Primary actor	Secondary	Stories
UC-12	Get recipe suggestions	Household user	AI services	US-4.01, 4.03, 4.04, 4.06, 10.03, 10.04
UC-13	Ask the cooking assistant	Household user	AI services	US-4.02, 4.07
UC-14	Mark meal as cooked	Household user	–	US-4.05
UC-15	Complete weekly pantry check	Household user	–	US-2.09, 10.01
UC-16	Update item status (used, thrown away)	Household user	–	US-2.08, 2.10, 2.11
UC-17	View household impact	Household user	–	US-9.01, 9.02, 9.03, 9.04, 9.11
UC-18	Get monthly waste insight	Household user	AI services	US-9.05
UC-19	Set a waste-reduction goal	Household user	–	US-9.06, 10.02
UC-20	Browse nearby surplus	Deal seeker	–	US-5.01, 5.02, 5.03, 5.06, 5.07
UC-21	Follow a store	Deal seeker	–	US-5.05, 11.03
UC-22	Reserve a bag	Deal seeker	–	US-5.04, 7.06
UC-23	Cancel a reservation	Deal seeker	–	US-7.04
UC-24	Rate store or report a problem	Deal seeker	–	US-5.08, 5.09
UC-25	Collect order with pickup code	Deal seeker, business staff	–	US-6.04, 7.02, 7.03, 11.04
UC-26	Process online payment or refund	Deal seeker	Payment provider	US-7.01, 7.05
UC-27	Create surplus listing	Business staff	–	US-6.01, 6.06
UC-28	Fill listing from a photo	Business staff	AI services	US-6.02
UC-29	Edit or cancel a listing	Business staff	–	US-6.05
UC-30	View business dashboard	Business staff	–	US-7.07, 9.07
UC-31	Screen listing automatically	Business staff	–	US-12.01
UC-32	Donate a bag or pantry item	Household user	–	US-8.05, 8.06
UC-33	Route food to a matched NGO	Scheduler	NGO coordinator	US-6.07, 8.02
UC-34	Set donation preferences	NGO coordinator	–	US-8.01
UC-35	Accept donation and book pickup	NGO coordinator	–	US-8.03, 8.04, 8.08
UC-36	Record meals distributed	NGO coordinator	–	US-8.07
UC-37	Sign up and sign in	App user	Authentication service	US-1.01, 1.02, 1.03, 1.04
UC-38	Manage profile and privacy	App user	–	US-1.05, 1.08, 10.06, 11.02
UC-39	Report a user or content	App user	–	US-12.02
UC-40	Manage staff accounts	Business owner	–	US-6.03
UC-41	Export CSR report	Business owner	–	US-9.08
UC-42	Register an organisation	Business owner, NGO coordinator	–	US-1.06, 1.07
UC-43	View NGO impact dashboard	NGO coordinator	–	US-9.09

ID	Use case	Primary actor	Secondary	Stories
UC-44	Verify an organisation	Administrator	–	US-12.03, 13.01
UC-45	Moderate flagged content	Administrator	–	US-13.02
UC-46	Resolve a dispute	Administrator	Payment provider	US-13.03
UC-47	Configure platform settings	Administrator	–	US-13.04
UC-48	Manage catalogue and recipes	Administrator	–	US-13.06
UC-49	View platform analytics	Administrator	–	US-13.05, 14.04, 14.05
UC-50	Export anonymised impact data	Government analyst	–	US-9.10

B.5 Key Use Case Descriptions

The five use cases below are written in the fully dressed format. They carry the main flows of the product: food entering the pantry, the marketplace handover, donation and impact.

UC-05 Scan receipt

Primary actor	Household user
Secondary actors	AI services (LLM through the AI gateway)
Stories	US-2.01, 2.02, 3.01, 14.01, 14.03
Preconditions	The user is signed in and belongs to a household.
Trigger	The user taps *Scan receipt* after shopping.
Main success scenario	1. The user photographs a paper receipt. 2. The app reads the text on the phone (ML Kit OCR) and masks names, phone numbers and card digits. 3. The app sends the masked text to the AI gateway. 4. The AI gateway asks the LLM to return each line as a pantry item (name, quantity, unit, category) and validates the result. 5. The system estimates an expiry date for each item (UC-10). 6. The app shows the items with confidence levels, highlighting low-confidence ones (UC-09). 7. The user corrects any item and taps *Confirm all*. 8. The system saves the items to the household pantry.
Alternative flows	**3a.** No internet: the photo is queued and processed when the phone reconnects. **4a.** The AI gateway times out (15 s) or returns invalid data: the app shows the OCR lines and the user adds items manually (UC-03). **7a.** The user deletes a wrongly read line; it is not saved.
Postconditions	The pantry holds the confirmed items, each with an expiry date and its source (printed or estimated). The user's corrections are logged without personal data (US-14.04).

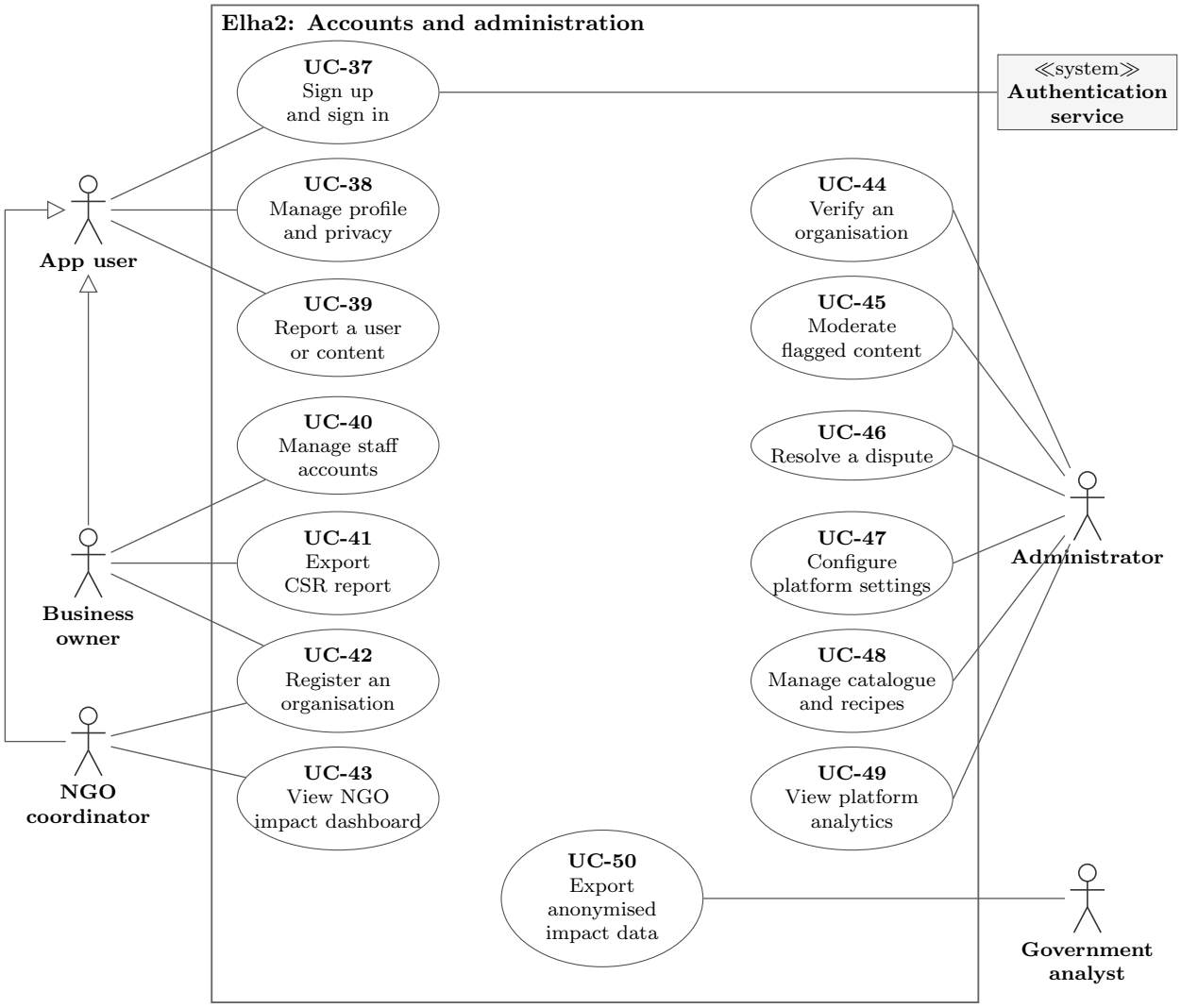


Figure B.5: Use case diagram 5: accounts, organisations, administration and reporting.

UC-22 Reserve a bag

Primary actor	Deal seeker
Secondary actors	Payment provider (only for online payment, UC-26)
Stories	US-5.04, 7.02, 7.06
Preconditions	The deal seeker is signed in; the listing is published and has quantity left.
Trigger	The deal seeker taps *Reserve* on a listing.
Main success scenario	1. The deal seeker chooses the quantity and the payment method (cash at pickup). 2. The system reduces the listing quantity in a single transaction. 3. The system creates the order with a QR code and a 4-digit pickup code. 4. The system confirms the reservation on screen and by push notification, and notifies the business.
Alternative flows	**1a.** The deal seeker chooses online payment: UC-26 runs before step 2. **1b.** The deal seeker has 3 no-shows in the last 30 days: reservation is refused for 7 days. **2a.** The last bag was taken a moment earlier: the system rejects the reservation and shows "sold out".
Postconditions	One bag is held for the deal seeker until the end of the pickup window; no bag is sold twice (NFR-06).

UC-25 Collect order with pickup code

Primary actors	Deal seeker, business staff
Stories	US-6.04, 7.03, 9.03, 11.04
Preconditions	The order is reserved and its pickup window is open.
Trigger	The deal seeker arrives at the store.
Main success scenario	1. The deal seeker shows the QR code in the app. 2. The staff member scans it, or types the 4-digit code, in the business portal. 3. The system checks that the code belongs to an open order for this branch. 4. The staff member hands over the bag and takes payment if it is cash. 5. The system marks the order *collected* and writes one verified impact record (kg, EGP, CO_2e).
Alternative flows	**3a.** The code is wrong or the order belongs to another branch: the system shows an error and nothing changes. ***a.** The window ends without a pickup: the order becomes a no-show and the bag is offered to an NGO (UC-33).
Postconditions	The order is closed, and the business and the buyer each see the same saving once in their dashboards.

UC-33 Route food to a matched NGO

Primary actor	Scheduler

UC-33 Route food to a matched NGO (continued)

Secondary actors	NGO coordinator
Stories	US-6.07, 8.01, 8.02, 8.04
Preconditions	The business has turned on auto-donation; at least one verified NGO serves the area.
Trigger	A listing's pickup window ends with unsold or uncollected quantity.
Main success scenario	1. The scheduler finds NGOs whose service area, accepted food types, capacity and hours match the food. 2. The system orders the matches, putting the business's default partner first and then the nearest. 3. The system sends a donation offer to the first NGO. 4. The NGO coordinator accepts and books a pickup time (UC-35). 5. The system notifies the business of the booked time. 6. At pickup, the business confirms the handover with the NGO's code, and one verified impact record is written.
Alternative flows	**4a.** No answer within 30 min, or the offer is declined: the system offers the food to the next NGO. **4b.** No NGO accepts before the food's safe-use time: the offer closes and the business is told it was not collected.
Postconditions	The surplus is booked for an NGO pickup, or the business knows that no NGO could take it.

UC-17 View household impact

Primary actor	Household user
Stories	US-9.01, 9.02, 9.03, 9.04
Preconditions	The user is signed in.
Trigger	The user opens the Impact tab.
Main success scenario	1. The system totals the user's verified impact records for this week, this month and all time. 2. The app shows kg of food saved, EGP saved, CO_2e avoided and meals provided, split by Prevent, Sell and Donate. 3. The app shows the monthly waste rate (food thrown away versus used) against the Egyptian average for the household's size. 4. The user taps *How we calculate* and sees the factors per food category, their source and their version date.
Alternative flows	**1a.** No verified actions yet: the app explains how impact is earned and links to the pantry and the deals. **3a.** Less than one month of pantry data: the waste rate is marked as provisional.
Postconditions	None; the use case only reads data.

Appendix C: To Be Determined List

ID	Open item	Where used	Close by
TBD-01	Which hosted LLM to use for receipt and voice parsing, chosen by accuracy and cost on our Egyptian test set	Section 2.2.1	Sprint 1

Appendix D: Verification and Traceability

This appendix states how each requirement will be verified, how the requirements were checked for quality, and which stakeholders each feature serves. The matrices are updated at the end of every sprint.

D.1 Verification Methods

Each requirement names one of four methods, and is accepted only when its acceptance test passes.

Method	Meaning
Test (T)	Automated or scripted execution with defined inputs and expected outputs (unit, integration, end-to-end, load or usability tests).
Demonstration (D)	The behaviour is shown working end to end at a sprint review, without detailed measurement.
Inspection (I)	Review of code, configuration, content or documents against a checklist.
Analysis (A)	Measurement or calculation against a stated target, such as AI accuracy on the labelled test set or uptime over a month.

Requirement group	Method	How it is verified
Functional requirements (US-1.01 to US-13.06) and business rules (BR-01 to BR-12)	Test, demonstration	Unit and integration tests for each acceptance criterion; end-to-end tests of the main flows (scan a receipt, reserve and collect a bag, auto-donate); demonstrated at each sprint review
AI features (stories marked **[AI]**, US-14.x, NFR-22, NFR-29, NFR-30)	Analysis, test	Accuracy measured on the labelled test set (US-14.02) in the CI pipeline at every release; fallback paths tested by simulating provider time-outs
Performance (NFR-01 to NFR-06)	Analysis, test	Load tests (e.g. k6) on a staging server; app start-up timed on a mid-range Android phone; concurrent-reservation test for overselling
Safety (NFR-07, NFR-08)	Test, inspection	Output-filter tests with unsafe prompts; licence and allergen checks on listings
Security and privacy (NFR-09 to NFR-15)	Inspection, test	OWASP checklist review; role-based access tests for every role; checks that photos and small groups are never exposed
Usability (NFR-16 to NFR-18)	Inspection	Accessibility audit with screen readers and contrast checkers; review of labels and Arabic wording
Reliability and availability (NFR-19 to NFR-23)	Analysis, test	Uptime monitoring; scheduler job logs; backup restore rehearsed once per release; offline mode tested in airplane mode

Requirement group	Method	How it is verified
Maintainability, portability and localisation (NFR-24 to NFR-28, NFR-31)	Inspection, analysis	Coverage reports; code reviews; device and browser test matrix; locale checks

D.2 Requirement Quality

Each requirement was checked against the qualities of a good requirement in ISO/IEC/IEEE 29148 [1], and the set as a whole against the qualities of a good requirements collection [2].

Quality	Meaning	How this SRS meets it
Each requirement		
Singular	One behaviour, no hidden conjunctions	Each requirement has one "shall" statement; details sit in its acceptance test
Feasible	Achievable with current technology and cost	Every AI capability uses an existing model or service (Section 2.2.1); the course scope fits our capacity (Section 6.6)
Unambiguous	Only one interpretation	Defined terms (Appendix A); vague words such as "fast" or "easy" are replaced by numbers
Complete	Measurable and fully describes the need	Each requirement carries a measurable constraint or an acceptance test
Consistent	Does not contradict another requirement	Reviewed within each feature and across the use case catalogue
Verifiable	Satisfaction can be established	Every requirement names a verification method and a test case (Section D.4)
Traceable	Uniquely identified, with clear links	Unique IDs, traced to use cases (Table B.1) and stakeholders (Section D.5)
Ranked	Prioritised	MoSCoW priority on every requirement
The set as a whole		
Complete	No missing functionality	14 features cover the full lifecycle, from sign-up to administration; 50 use cases cover every actor
Consistent	With itself and with applicable documents	Checked against the PDPL and NFSA rules
Bounded	Within the scope of the system's intent	Every requirement serves the mission (Section 1.3.2); out-of-scope items are listed in Section 1.3.3
Affordable	Satisfiable within budget	Free tiers and capped AI cost (US-14.05); the course scope is limited to 146 story points

D.3 Validation Matrix

Business need and safety impact are rated High, Medium or Low; volatility is how likely the requirements are to change.

Feature	Business need	Volatility	Safety impact	Validation approach
E1 Accounts and onboarding	High	Low	Medium	Prototype walkthrough with household users
E2 Automated smart pantry	High	High	Low	Prototype and usability test with real receipts
E3 Expiry intelligence	High	Medium	Medium	Shelf-life table reviewed against food-storage guidance
E4 Recipe assistant	Medium	Medium	Medium	Recipe and safety-note review by the team
E5 Surplus marketplace	High	Medium	Low	Interviews with deal seekers (students)
E6 Business portal	High	Medium	Medium	Interviews with a café and a bakery
E7 Orders and payments	High	Medium	Low	Walkthrough of reservation and pickup with a business
E8 NGO donation	High	Low	Medium	Walkthrough with an NGO coordinator
E9 Impact and reporting	High	Medium	Low	Review of the calculation method
E10 Engagement and Ramadan	Low	High	Low	Household survey
E11 Notifications	Medium	Low	Low	Prototype review
E12 Trust and safety	High	Medium	High	Review against NFSA rules and the risk register
E13 Administration	Medium	Low	Medium	Team walkthrough
E14 AI platform	High	High	Medium	Accuracy report on the labelled test set

D.4 Verification Matrix

Every requirement is linked to its method and test case. *Status* is updated at every sprint review (Not run, Passed or Failed). Methods: T = Test, D = Demonstration, I = Inspection, A = Analysis.

Req.	M	Test case	Status	Req.	M	Test case	Status
US-1.01	T	TC-1.01	Not run	US-9.04	T	TC-9.04	Not run
US-1.02	D	TC-1.02	Not run	US-9.05	A	TC-9.05	Not run
US-1.03	D	TC-1.03	Not run	US-9.06	T	TC-9.06	Not run
US-1.04	T	TC-1.04	Not run	US-9.07	D	TC-9.07	Not run
US-1.05	T	TC-1.05	Not run	US-9.08	T	TC-9.08	Not run
US-1.06	D	TC-1.06	Not run	US-9.09	D	TC-9.09	Not run
US-1.07	D	TC-1.07	Not run	US-9.10	T	TC-9.10	Not run
US-1.08	T	TC-1.08	Not run	US-9.11	I	TC-9.11	Not run
US-2.01	T	TC-2.01	Not run	US-10.01	T	TC-10.01	Not run
US-2.02	D	TC-2.02	Not run	US-10.02	T	TC-10.02	Not run
US-2.03	T	TC-2.03	Not run	US-10.03	T	TC-10.03	Not run
US-2.04	T	TC-2.04	Not run	US-10.04	A	TC-10.04	Not run
US-2.05	A	TC-2.05	Not run	US-10.05	I	TC-10.05	Not run
US-2.06	A	TC-2.06	Not run	US-10.06	T	TC-10.06	Not run

Req.	M	Test case	Status	Req.	M	Test case	Status
US-2.07	D	TC-2.07	Not run	US-11.01	T	TC-11.01	Not run
US-2.08	T	TC-2.08	Not run	US-11.02	T	TC-11.02	Not run
US-2.09	T	TC-2.09	Not run	US-11.03	T	TC-11.03	Not run
US-2.10	T	TC-2.10	Not run	US-11.04	T	TC-11.04	Not run
US-2.11	D	TC-2.11	Not run	US-11.05	D	TC-11.05	Not run
US-3.01	T	TC-3.01	Not run	US-12.01	T	TC-12.01	Not run
US-3.02	A	TC-3.02	Not run	US-12.02	T	TC-12.02	Not run
US-3.03	T	TC-3.03	Not run	US-12.03	T	TC-12.03	Not run
US-3.04	I	TC-3.04	Not run	US-12.04	T	TC-12.04	Not run
US-3.05	D	TC-3.05	Not run	US-13.01	D	TC-13.01	Not run
US-3.06	D	TC-3.06	Not run	US-13.02	D	TC-13.02	Not run
US-4.01	T	TC-4.01	Not run	US-13.03	T	TC-13.03	Not run
US-4.02	A	TC-4.02	Not run	US-13.04	T	TC-13.04	Not run
US-4.03	T	TC-4.03	Not run	US-13.05	D	TC-13.05	Not run
US-4.04	T	TC-4.04	Not run	US-13.06	D	TC-13.06	Not run
US-4.05	T	TC-4.05	Not run	US-14.01	T	TC-14.01	Not run
US-4.06	D	TC-4.06	Not run	US-14.02	A	TC-14.02	Not run
US-4.07	T	TC-4.07	Not run	US-14.03	T	TC-14.03	Not run
US-5.01	T	TC-5.01	Not run	US-14.04	I	TC-14.04	Not run
US-5.02	T	TC-5.02	Not run	US-14.05	T	TC-14.05	Not run
US-5.03	D	TC-5.03	Not run	NFR-01	T	TC-N01	Not run
US-5.04	T	TC-5.04	Not run	NFR-02	T	TC-N02	Not run
US-5.05	T	TC-5.05	Not run	NFR-03	T	TC-N03	Not run
US-5.06	A	TC-5.06	Not run	NFR-04	T	TC-N04	Not run
US-5.07	D	TC-5.07	Not run	NFR-05	A	TC-N05	Not run
US-5.08	T	TC-5.08	Not run	NFR-06	T	TC-N06	Not run
US-5.09	T	TC-5.09	Not run	NFR-07	T	TC-N07	Not run
US-6.01	T	TC-6.01	Not run	NFR-08	I	TC-N08	Not run
US-6.02	A	TC-6.02	Not run	NFR-09	I	TC-N09	Not run
US-6.03	T	TC-6.03	Not run	NFR-10	T	TC-N10	Not run
US-6.04	T	TC-6.04	Not run	NFR-11	I	TC-N11	Not run
US-6.05	T	TC-6.05	Not run	NFR-12	T	TC-N12	Not run
US-6.06	T	TC-6.06	Not run	NFR-13	I	TC-N13	Not run
US-6.07	T	TC-6.07	Not run	NFR-14	T	TC-N14	Not run
US-7.01	T	TC-7.01	Not run	NFR-15	T	TC-N15	Not run
US-7.02	T	TC-7.02	Not run	NFR-16	I	TC-N16	Not run
US-7.03	T	TC-7.03	Not run	NFR-17	I	TC-N17	Not run
US-7.04	T	TC-7.04	Not run	NFR-18	I	TC-N18	Not run
US-7.05	T	TC-7.05	Not run	NFR-19	T	TC-N19	Not run
US-7.06	D	TC-7.06	Not run	NFR-20	D	TC-N20	Not run
US-7.07	T	TC-7.07	Not run	NFR-21	T	TC-N21	Not run
US-8.01	T	TC-8.01	Not run	NFR-22	T	TC-N22	Not run
US-8.02	T	TC-8.02	Not run	NFR-23	A	TC-N23	Not run
US-8.03	T	TC-8.03	Not run	NFR-24	A	TC-N24	Not run
US-8.04	T	TC-8.04	Not run	NFR-25	I	TC-N25	Not run
US-8.05	T	TC-8.05	Not run	NFR-26	T	TC-N26	Not run
US-8.06	D	TC-8.06	Not run	NFR-27	T	TC-N27	Not run
US-8.07	I	TC-8.07	Not run	NFR-28	I	TC-N28	Not run

Req.	M	Test case	Status	Req.	M	Test case	Status
US-8.08	T	TC-8.08	Not run	NFR-29	A	TC-N29	Not run
US-9.01	T	TC-9.01	Not run	NFR-30	D	TC-N30	Not run
US-9.02	I	TC-9.02	Not run	NFR-31	I	TC-N31	Not run
US-9.03	T	TC-9.03	Not run				

D.5 Stakeholder Matrix

The matrix shows which stakeholder class each feature's requirements are written for (• = at least one requirement written for that class; ◦ = a requirement written for any app user).

Feature	House-hold	Deal seeker	Busi-ness	NGO	Admin	Gov. analyst	Team
E1 Accounts and onboarding	•	◦	•	•			
E2 Automated smart pantry	•						
E3 Expiry intelligence	•						
E4 Recipe assistant	•						
E5 Surplus marketplace		•					
E6 Business portal			•				
E7 Orders and payments	◦	•	•				
E8 NGO donation	•	◦	•	•			
E9 Impact and reporting	•	◦	•	•		•	•
E10 Engagement and Ramadan	•	◦					
E11 Notifications	◦	•					
E12 Trust and safety	◦	•					•
E13 Administration					•		
E14 AI platform	•						•

Appendix E: Declaration of AI Use

The requirements in this document were defined by our team, building on the concept, features and research from our Phase 0 ideation work. We used Claude (Anthropic) to help structure and draft the backlog, the use case diagrams and this document, and to prepare its LaTeX layout. All content was reviewed and edited by us, and we take full responsibility for the final document.